DEPARTMENT OF MANAGEMENT SCIENCE

University of Strathclyde Business School

Human rights watch

Albinism in Africa

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MSc Data Analytics

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DECLARATION

Except where explicitly stated, all the work in this dissertation – including any appendices – is my own and was carried out by me during my MSc course. It has not been submitted for assessment in any other context.

Signed: Elise L'Hospitalier

EXECUTIVE SUMMARY

Albinism is a genetic condition that can affect skin, hair and eye colour of the persons impacted. Besides the physical consequences of albinism, persons with albinism face multiple challenges during their lives including vision impairments and skin more sensitive to the sun. Moreover, persons with albinism face discrimination during their entire lives: at school when they are children, while looking for work, when needing healthcare, when needing to access justice, etc. In Africa especially, where persons with albinism are particularly visible, persons with albinism often face human rights violations. Every day, their lives are threatened. They are often kidnapped, abducted or killed for various reasons by strangers, members of their communities or even members of their families.

The project's purpose was to build a human rights observatory for persons with albinism in Africa. Multiple organisations have built online platforms to talk about the issue and those platforms were carefully analysed. It was concluded that no existing online platform provided visualisations of all available data on human rights violations of persons with albinism in Africa. Therefore, its aim was to display cases of human rights violations against persons with albinism in Africa. Users of the platform were identified as non-governmental organisations, policy makers, human rights researchers and persons with albinism. The platform that was built is available at this link: Home Page · Streamlit

An automated data collection pipeline was designed and implemented. Over one hundred documents freely available online were automatically analysed and more than 500 excerpts mentioning human rights were extracted from the documents, using gemini 3.5-Flash model. Multiple natural language processing methods were then used to extract meaningful information from each excerpt, such as human rights topics or countries. The dataset was carefully cleaned, before the platform could be built.

Various functions were integrated into the platform to fill its purpose, such as an interactive map of Africa and a search engine allowing the user to search incidents by keyword. To ensure the functions that were built were well-integrated into the platform, the platform was then tested by multiple users. Respondents performed key tasks on the platform and rated the ease with which they performed each task before answering questions on the general functioning of the platform. The general feedback was that the platform functioned well and the various functions were well-integrated. Few areas of improvement were identified, and modifications were made based on these areas of improvement.

The platform build is an efficient first version and working on it during three months was a great learning experience.

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INTRODUCTION

1 SCENE SETTING

Albinism is a genetic condition resulting in a lack of pigmentation in hair, skin and/or eyes. It affects people regardless of gender, skin colour or ethnicity. The worldwide prevalence of oculocutaneous albinism, the most common form of albinism (Durojaye and Nabaneh, 2019) is reportedly around 1:17000 (Wright et *al.*, 2015) but that number may be unreliable (Ero *et al.*, 2019). In Africa, the prevalence of albinism is higher: it usually is between 1:15000 and 1:5000 depending on the population studied (Wright et *al.*, 2015; Ero *et al.*, 2019). However, it can get as high as 1:1000 in some populations, such as a Tonga tribe in Zimbabwe (Lund *et al.*, 1997).

People with albinism are prone to having vision impairments and are much more likely to develop skin cancer, as their skin is sensitive to solar ultraviolet radiations (Tambala-Kaliati, Adomako and Frimpong-Manso, 2021 ; Wright et *al.*, 2015). The high sun exposure in sub-Saharan Africa is therefore a danger to persons with albinism living there, especially as people in sub-Saharan Africa often live in rural areas and spend quite a lot of time outdoors (Wright et *al.*, 2015). Besides, persons with albinism in Africa have a shorter life expectancy than normal. For example, almost 90% of persons with albinism are under 30 years old in Nigeria (Durojaye and Nabaneh, 2019) while the life expectancy in Nigeria is 64.3 years old (World Health Organisation, 2026).

Albinism has been implicitly recognized as a disability by the Committee on the Rights on Persons with Disabilities. Here, disability is not to be understood from a medical perspective – where disabled persons would be seen as patients that need to be healed, but rather from a human rights or social approach. In this perspective, a disability is an impairment that would exclude someone from social life (Durojaye and Nabaneh, 2019). The Committee on the Rights of Persons with Disabilities, in communication 023/2014 for the Y v United Republic of Tanzania case recognized that Tanzania had violated multiple articles of the convention when the police forces did not make it possible to identify and condemn the perpetrators of violent attacks against a person with albinism(Convention on the Rights of Persons with Disabilities, 2018).

The Universal Declaration of Human Rights (1948) states: “All human beings are born free and equal in dignity and rights”. However, persons with albinism are often the victims of human rights violation (Ero *et al.*, 2019). Because of these violations, the United Nations decided, in 2015, to appoint an Independent Expert on the enjoyment of human rights by persons with

albinism. The report *People with albinism worldwide – a human rights perspective* (Ero *et al.*, 2019), focuses on 6 human rights issues faced by persons with albinism: right to life, right to education, right to work and adequate standard of living, access to justice, right to health and multiple and intersecting discriminations. These six points were used for this project.

Multiple and intersecting discrimination

Persons with albinism in the world face multiple forms of discriminations: discrimination based on their disability (as mentioned before) and based on their skin colour. In Africa, and especially in Sub Saharan Africa, persons with albinism are particularly visible and they face even more discriminations from their peers. In this region, persons with albinism have to endure name calling, exclusion from social life in every setting – when shopping, at school, at work, etc. (Ero *et al.*, 2019; Tambala-Kaliati, Adomako and Frimpong-Manso, 2021).

Right to life

The Right to life is described in article 3 of the Universal Declaration of Human Rights as the "right to life, liberty and security of person" (United Nations, 1948).

The physical integrity of people with albinism in Africa is often compromised. Persons with albinism are the targets of violent attacks and killings in multiple countries in Africa such as Mozambique (Muscati, no date), Malawi (Tambala-Kaliati, Adomako and Frimpong-Manso, 2021), Tanzania (Convention on the Rights of Persons with Disabilities, 2018). These attacks are often due to several myths and legends about albinism that are widely believed in in Africa. It is, for example, believed that body parts of people with albinism bring good luck or good fortune or that having intercourse with a woman or girl with albinism can cure HIV. Persons with albinism might also be accused of witchcraft (Ero, 2016). In Africa, between 2010 and 2020, over 700 persons with albinism have been killed, mutilated or traumatized due to accusations of witchcraft (Ero *et al.*, 2019).

Right to education

The article 26 of the UDRH states that every child has the right to free education. School should be available, accessible and compulsory for all children (United Nations, 1948).

Children with albinism are often deprived of their right to education. Because of their vision impairments, they need special modified textbooks (with a larger font size) or visual aids. Teachers in Africa and everywhere in the world are often not aware of these special needs (Tambala-Kaliati, Adomako and Frimpong-Manso, 2021; Muscati, no date).

Moreover, teachers and classmates of persons with albinism may discriminate against them, calling them names and forcing them to drop out of school (Tambala-Kaliati, Adomako and

Frimpong-Manso, 2021). The threat of their child getting kidnapped while going to school may also force parents to keep their child at home instead of allowing them to get an education, leading to them being illiterate.

Right to work and adequate standard of living

The article 23 of the UDHR states that everyone has the right to work in fair conditions, to freely choose where they want to work and to earn a fair salary from their work (United Nations 1948).

The right to an adequate standard of living is described in article 25 of the Universal Declaration of Human Rights. It states that every human being should have access to clothes, food, housing and medical care when necessary (United Nations, 1948).

Right to health

The right to health is not explicitly mentioned in the Universal Declaration of Human Rights but is part of the right to an adequate standard of living that was mentioned previously. It has been described in the International Covenant on Economic, Social and Cultural rights as "the right of everyone to the enjoyment of the highest attainable standard of physical and mental health." (United Nations, 1966). This implies that healthcare should be available, accessible, acceptable and of quality. Healthcare should be accessible to every human being, both physically and financially and it should cater to everyone, with respect to their culture.

The skin of persons with albinism is highly sensitive to the sun and needs special sunscreen lotion. Persons with albinism often do not know about this special need, and when they are made aware of it, it is not always easily accessible (Tambala-Kaliati, Adomako and Frimpong-Manso, 2021).

Access to justice

Article 8 of the UDHR mentions that "Everyone has the right to an effective remedy by the competent national tribunals for acts violating the fundamental rights granted him by the constitution or by law" (United Nations, 1948). The Convention on the Rights of Persons with Disabilities also mentions in article 13 that the necessary adaptations should be made to accommodate the needs of persons with disabilities, which persons with albinism have been classified as (United Nations, 2006).

Currently, people with albinism often struggle to access justice when their rights have been violated. That is even more true for women and children (Ero *et al.*, 2019). The Committee on the Rights of Persons with Disabilities has recognized that the government of Tanzania was guilty of not properly investigating the attacks against a person with albinism. This report also

mentions that a majority of the murders committed on persons with albinism since 2020 have not resulted in a prosecution (2018). The myth that persons with albinism “do not die, but simply disappear” also discourages communities from looking to investigate the death of persons with albinism in Africa (Ero, 2016).

The aim of this project was to build a human rights observatory for persons with albinism in Africa.

2 PROJECT PURPOSE AND PLAN

The project was proposed by academic superior Catherine Chavula. Its aim was to build a human rights observatory for persons with albinism in Africa.

The plan for the project was broadly defined at the beginning of the project. Broad steps, timing of meetings with academic supervisor and deadlines were defined with academic supervisor at the beginning of the project. This plan slightly evolved throughout the project, as the methodology was defined. This plan is available on Chart 1.

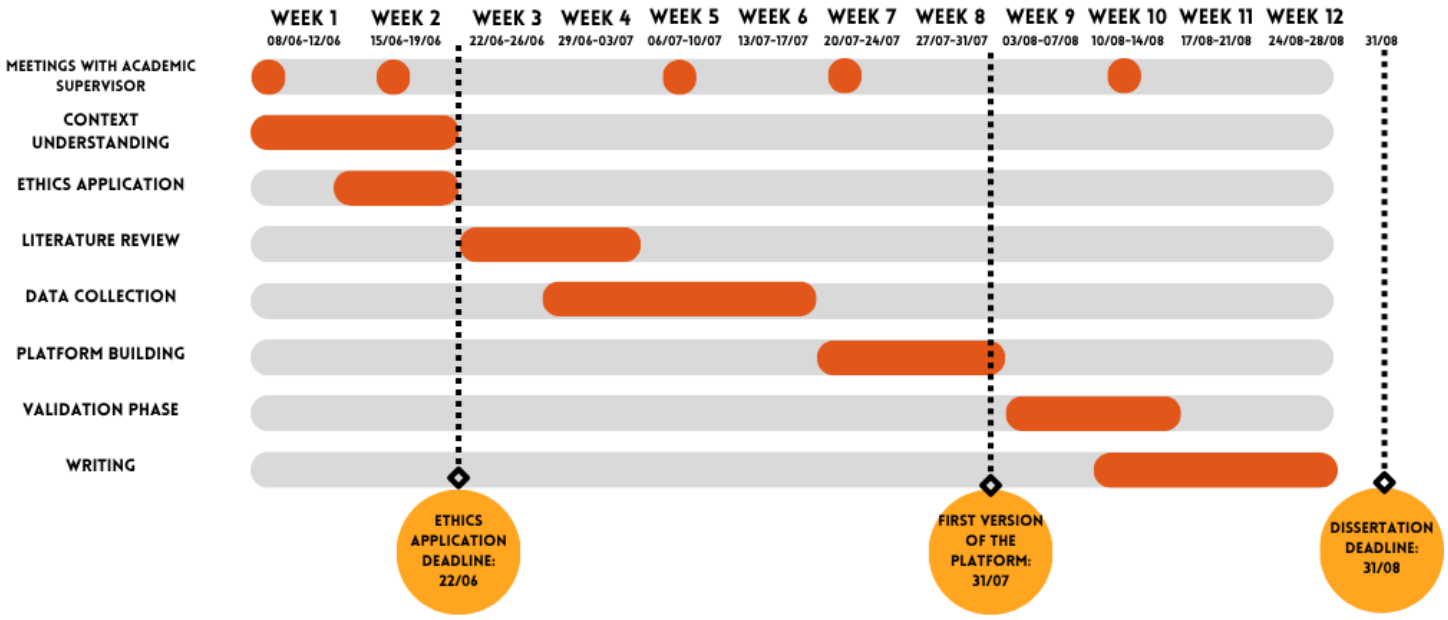


Chart 1: Initial project plan

Chart 2 shows the details of what was done each week of the project, in comparison with the initial plan.

	Week 1	**Week 2**	**Week 3**	**Week 4**	**Week 5**	**Week 6**
Plan	Context understanding	Context understanding, ethics application	Literature review	Literature review, Data collection	Data collection	Data collection
Reality	10/06: Initial meeting with academic supervisor Reading of paper on albinism	Reading papers on albinism 16/06: meeting to discuss ethics application Draft of the platform and first submission	Rejection of ethics application, re-submission of ethics application	Finished draft of the envisioned platform Started literature review for data collection	Finished literature review 06/07: meeting to discuss data collection	Test of each component of the data collection pipeline

		of ethics application		and platform building	Built data collection pipeline	

	Week 7	**Week 8**	**Week 9**	**Week 10**	**Week 11**	**Week 12**
Plan	Platform building	Platform building	Validation phase	Validation phase	Dissertation writing	Dissertation writing
Reality	Started data collection 21/07: meeting to discuss the data collection pipeline	Finished data collection process while building the platform	First version of the platform was proposed to supervisor, was modified based on the feedback that was received Writing of literature review	12/08: meeting to discuss final platform and validation process Validation process Writing of methodology part	Dissertation writing	Dissertation writing

Chart 2: Advancement of the project

DEPARTMENT OF MANAGEMENT SCIENCE

University of Strathclyde Business School

Client report

MSc dissertation: Human rights watch – albinism in Africa

Elise Marie Hélène L'Hospitalier

Academic supervisor: Catherine Chavula

University of Strathclyde

2026

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CHAPTER 1 SUMMARY

Persons with albinism in Africa are often victims of cruel acts such as killing, abduction or kidnapping. They face discriminations and human rights violations during their entire lives. Multiple non-governmental organisations support them and the United Nations have appointed an Independent expert on the enjoyment of human rights by persons with albinism.

Human rights violations are documented online. NGOs and multiple medias often publishes articles and reports on violations of human rights of persons with albinism. There is currently no platform available online summarising all of this data. The aim of this project was to build a human rights observatory for persons with albinism in Africa.

The final version of the platform that was built is available at this link: Home Page · Streamlit

It provides visualisations of human rights violations instances against persons with albinism in Africa. An extensive database was created to build this app. Documents available online were analysed to build this dataset.

This report gives details about the steps taken to build the human rights platform. The first chapter reviews the current works and available solutions for the platform. The second chapter presents the approach that was taken to build the platform and the details of the implementation. The third step details the results that were obtained, giving an overview of the dataset that was built and showing details of the main functions implemented in the app. The fifth chapter discusses the results and interpretations of the results and highlights the limitations and possible future steps of the project.

CHAPTER 2 BACKGROUND AND LITERATURE REVIEW

This section reviews the current literature relevant to the project. It is organised in three parts. The first part goes through current existing platforms about persons with albinism, reviewing the common features included in them. The second part reviews what is natural language processing, with a special focus on natural language processing techniques that were used throughout the project. Finally, the third part reviews information retrieval techniques.

2.1 Existing platforms

The first step of the dissertation was to explore existing platforms about human rights of persons with albinism and other human rights issues, in order to have an overview of what is currently working.

2.1.1 Existing platforms for human rights of persons with albinism in Africa

People with albinism face multiple human rights violations on a daily basis everywhere in the world. Multiple organisations have documented these violations and have built online platforms to share them. These platforms were most of the time built by non-governmental organisations (NGOs), to inform the public about albinism and their actions. These organisations are present worldwide or specifically in Africa or regions of Africa.

Four platforms were analysed: Standing voice (*Standing Voice*, no date), Under the Same Sun – Albinism news and stories ('Albinism News & Stories', 2026), Global Albinism Alliance – human rights news (Global Albinism Alliance, no date) and Meet the champions of the albinism cause – United Nations (United Nations, no date).

Many of these platforms contain stories and news about persons with albinism, sometimes organised by themes, such as the Standing Voice platform. Other platforms, such as the Under the Same Sun platform allow the user to search articles and stories using a search engine. These stories and articles often focus on actions that were carried out by the organisation. The Human Rights News platform from the Global Albinism Alliance also contains the stories of persons with albinism worldwide. Each story is detailed, long and available in multiple languages. The United Nations human rights office's platform on albinism also includes stories of persons with albinism worldwide, mainly focused on outstanding achievements from persons with albinism.

2.1.2 Existing platforms surrounding other human rights issues

Plan International's Girls' rights database (Plan international, no date) is a platform for girls' rights and gender equality advocates, allowing users to search through an extensive database - more than 10 000 documents. Multiple filters are also available to enable the user to easily

find any document, based on the type of document, the topics covered in the documents, the year during which the document was published.

The Mapa de ataques a Defensoras Ambientale (map of attacks against women defending the environment), jointly created by CEDIB (Centro de Documentatión e Informatión Bolivia) and CONTIOCAP (Coordinadora Nacional de Defensa de Territorios Indígenas Originarios Campesinos y Áreas Protegidas) (CEDIB-CONTIOCAP) collects and shows on a world map attacks against environment defender. The colour coding indicates how many attacks have been reported, and when clicking on a specific location, details about the reported attacks can be seen.

These two platforms are complimentary and provided useful insights and starting points for the human rights albinism in Africa platform. A map is a quick way of visualising spatial trends and filtering the data based on a region of interest. However, this does not allow the user to simply search documents, and it can make it hard to find specific documents. The search engine, on the contrary, makes it quick and easy to use multiple filters at the same time and look for a specific document but lacks data visualisation tools.

2.2 Natural language processing

To build the platform, a way to analyse and extract information from documents available online had to be found. This was done using multiple natural language processing techniques.

2.2.1 Definition and overview of natural language processing

Natural language processing (NLP) is a field of artificial intelligence that was first developed in the 1950s, using grammar and syntax-based rules, to carry out translation tasks (Supriyono et *al*., 2024; Khurana et al., 2023). The field of Natural Language Processing are intended to enable machines to process, interpret and generate human written text, which is unstructured data (Rayhan, Kinzler and Rayhan, 2023). Current uses of NLP techniques are varied and include language translation, social media analysis through sentiment analysis tools, legal document summarisation (Rayhan, Kinzler and Rayhan, 2023).

Natural language processing techniques can be divided into two categories: Natural language understanding - analysing human text to extract information - and Natural language generation – producing human text (Khurana et *al*., 2023).

Natural language understanding techniques aim at understanding all subtleties that can be included in human text, such as irony or sarcasm, nuances, humour and abstract concepts (Cambria, 2025). To do so, NLU algorithms have to understand meaning in human language at different linguistic levels: syntactic level (analysing the grammatical structure of a group of words), semantic level (understanding the actual meaning of each work in a group of words),

discourse (analysing the logical structure between group of words), pragmatic level (understanding what is implied in the text) (Khurana et *al*., 2023; Bates, 1995). Natural language understanding has multiple applications in various fields: in finance, predicting market trends; in medicine, offering early diagnostic by analysing medical notes; in the legal field, helping experts navigate the high number of documents, and making sure the law is correctly applied (Han *et al.*, 2025).

Natural language processing is still a relatively new field and has been evolving quickly in the last decades. Early NLP techniques (1950s and even before) largely relied on predefined rules. Later, machine learning techniques were introduced and used for natural language processing, before deep learning algorithms became more popular (Johri et *al*., 2021).

BERT models

BERT (Devlin *et al.*, 2019) is a pre-trained Transformer based language model, which uses bidirectional learning and masked language modelling. BERT is pre-trained on unlabelled data first, for multiple tasks such as masked language modelling (masking a work in the sentence, and training the model to predict it, using bidirectional learning) and Next Sentence Prediction (predicting whether a sentence follows another or not) (BERT 101 - State Of The Art NLP Model Explained, 2024; Gardazi et *al*., 2025).

He et *al*. (2021) present an enhanced version of the BERT algorithm: DeBERTa, Decoding Enhanced BERT with disentangled attention. The main improvement of DeBERTa over BERT comes from the disentangled attention mechanism and the enhanced masked decoder. With DeBERTa, the content and position of a token are represented as two different vectors (instead of one for BERT), and attention weights are computed separately for each vector. To enhance the masked decoder's performances, DeBERTa also uses the absolute positions of the context words as inputs (He et *al*., 2021).

BERT models can fine-tuned to conduct a variety of tasks: grammatical error detection, named-entity recognition, machine translation or zero-shot classification (Gardazi et *al*., 2025).

2.2.2 Zero-shot classification

Zero-shot text classification is an NLP task which intends to classify texts into categories that were not specified during the model training (Hugging Face, 2023). Once the model has been trained, it takes as an input the text to classify and candidate labels and returns a probability score for every candidate label.

Laurer et *al*. (2024) trained zero shot classifier models based on several pre-trained transformer models. They fine-tuned the models using multiple NLI datasets (mentioned before), to ensure that many different categories were seen during training. They concluded

that the model trained on DeBERTaV3 was the one that performed best, based on the balanced accuracy.

2.2.3 Named-entity recognition

Named-entity recognition is a natural language processing technique that aims at recognising specific entities, such as city names, company names or dates in a text (Kalyani, 2025). Several models have been developed for named entity recognition, such as BERT, Reinforcement learning or fine-tuned large language models. Named entity recognition has multiple applications in different fields: extracting insights from medical documents, extracting legal entities and codes from large legal documents, extracting vendor names from financial transactions, etc (Kalyani, 2025).

2.3 Information retrieval techniques

Information retrieval algorithms can be divided into different categories: lexical search (based on exact keyword matching), semantic search (methods that include semantic similarity between words) and hybrid search method, that combine lexical and semantic search (Gabryel, Kocić and Gabryel, 2026).

The most popular lexical search algorithm is the BM25 algorithm (best matching 25). BM25 uses term frequency weighting (TF-IDF) (Gabryel, Kocić and Gabryel, 2026). Term frequency refers to how frequent a word in the query is in the text: more mentions of an item suggest that the text is relevant to the query. Inverse document frequency refers to how rare a keyword is across all documents in the corpus: if a keyword is consistently present across all documents, it should carry less weight in the query (Zawanah, 2025).

This algorithm is efficient and quick to use but has a major limitation: if a user uses a synonym of a word, the algorithm cannot identify this. To overcome this limitation, semantic search algorithm can be used (Gabryel, Kocić and Gabryel, 2026). Semantic search algorithms encode both the query and the documents of the corpus, using bi-encoder models (Gabryel, Kocić and Gabryel, 2026; Farivar, 2025).

This chapter has reviewed existing online platforms surrounding human rights violations of persons with albinism, current existing natural language processing techniques and methods to extract meaningful insights from documents. The following chapter presents the methodology adapted during the project to design, build and test the platform.

CHAPTER 3 METHODOLOGY

The aim of this project is to build and evaluate a human rights observatory for persons with albinism in Africa. This human rights observatory will serve as a data visualisation platform, allowing users to view and filter through human rights violations against persons with albinism in Africa. To achieve this purpose, a four-steps approach was taken. This section details this approach.

This section contains five parts. The first part introduces the project methodology and the steps that were taken. The next four parts detail each of the four steps of the project: defining the scope and requirements of the platform, designing the platform, implementing it and finally, testing it.

3.1 Approach

The project adopted a waterfall model. A waterfall model is a simple model often used in software development (Saravanos, 2025). It is commonly made of five distinct phases: requirements analysis, design of the system, implementation, testing and release and maintenance. This approach was followed as it was a simple and straightforward approach, while still being robust. For this project, the first four phases of this model were followed. Figure 1 shows the steps that were followed during the project. The methodology of each of the steps is described in the rest of this section.

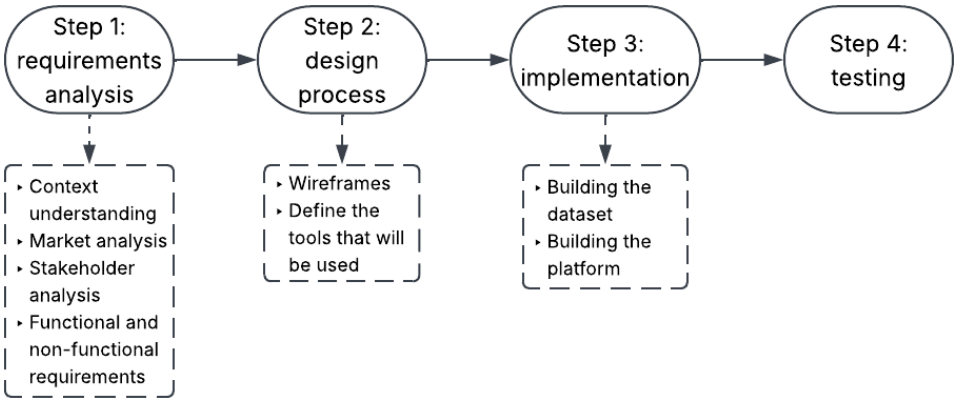


Figure 1: Steps followed in the project, based on the Waterfall model

3.2 Requirements analysis

Requirements analysis is the analysis of the needs of the platform, which features will have to be included and how the software will perform (Saravanos, 2025). Future users of the platform are considered during this phase. For this project, this phase included four steps: reading articles about the problem in order to fully understand the context, conducting a market

analysis to compare existing platforms and identify gaps, conducting a stakeholder analysis and building stakeholder personas and, finally, defining functional and non-functional requirements for the platform.

3.2.1 Context setting

In order to build a human rights platform showing human rights violations of persons with albinism, the first step was to get information about albinism and the situation of persons with albinism in Africa. This was done through reading scientific papers, news articles and browsing through multiple web pages of organisations. The aim of this first step was to explore the key issues; the existing approaches and the gaps associated with these approaches.

Myths about albinism have existed for a long time, at first to compensate a lack of knowledge about the causes of the condition. Persons with albinism face discrimination because of these myths, particularly in African countries, where they are especially visible (Ero, 2016). The first report of the International Expert on the Enjoyment of Human Rights by Persons with Albinism classified human rights violations against persons with albinism in six categories – six different fundamental human rights: right to life, right to education, right to work, access to justice, right to health and adequate standards of living (Ero *et al.*, 2019). This classification was an important design insight, as these six categories provided an interesting structure for organising the information on the platform. These categories were therefore incorporated in the platform to help the user explore the information, in a logical way.

3.2.2 Market analysis

A market analysis was then conducted to have an overview of the existing platforms which have characteristics similar to those of the envisioned platform. The strengths and weaknesses of each analysed platform were identified. Table 1 summarises the strengths and weaknesses of the platforms that were analysed.

Platform	Strengths	Weaknesses
Standing voice (*Standing Voice*, no date)	Complete website with a lot of information. It is easily navigable and contains a news / stories tab.	It is hard to find specific bits of information, their sources are not mentioned clearly.
Under the Same Sun – Albinism news and stories ('Albinism News & Stories', 2026)	Very complete website, that contains quite a lot	Focused only on the stories from their communities, which

	of information, focused on advocacy	introduces bias, all the information is written, there are barely any visualisations
Global Albinism Alliance – human rights news (Global Albinism Alliance, no date)	There are plenty of articles written on the platform, which are very informative.	It can be quite long to read everything in full to get the information that is needed.
Meet the champions of the albinism cause – United Nations (United Nations, no date)	Well organised, with the first page of the website showing stories from persons with albinism	Everything on the website is just text (and pictures), there are no highlighted sections
Girl's rights platform (Plan international, no date)	It is based on an extensive dataset of more than 10 000 documents collected	The design can be quite rebarbative to new users.
Mapa de ataques a Defensoras Ambientale (CEDIB-CONTIOCAP)	The map is very informative and easy to use	The filters are not intuitive to use

Table 1: Market analysis

Multiple platforms' aim is to inform people on albinism, i.e., give definitions, explain the difficulties faced by persons with albinism. A lot of platforms are built by one specific NGO and tackle the problem from an organisation's point of view (*Standing Voice*, no date; Albinism News & Stories, 2026). A major limitation of these platforms is that they focus on the work and actions carried out by these organisations. They lack an independent point of view on the human rights concerns of persons with albinism.

United Nation reports provide a more global point of view, as they work with multiple organisations everywhere in the world (Ero *et al.*, 2019). This information is presented in long, static PDFs. There is, however, no existing platform to visualise data on human rights violations against people with albinism that also provides a global point of view. This creates a need for an interactive platform that can be used to explore and understand these human rights issues across the African countries.

The two last platforms on Table 1, the Girls' Rights platform and the Mapa de ataques a Defensoras Ambientale are platforms on other human rights issues. The features on these platforms are interesting for the platform design process. They have features

that work really well: the searching engine on the Girls' Rights platform (Plan international, no date) and the map of the environment defenders' platform (CEDIB-CONTIOCAP) are particularly interesting and have been used as inspiration for the platform design.

3.2.3 Stakeholder analysis

The next step for the platform design was conducting a stakeholder analysis. Based on the market analysis that was conducted, multiple stakeholders were identified and stakeholders' personas were established. Dumouchel *et al.* (2020) described stakeholders' personas as "user archetypes". For each of the personas they created for their platform, they included a backstory, pain points and goals with the platform. Making these personas as human and realistic as possible is a key is the platform building process because it creates real persons for the UX designer. They can then emphasise with the personas and build the best platform possible for them (Dykes, 2025).

An approach similar to Dumouchel *et al.* (2020) was taken to conduct stakeholder analysis for the envisioned platform. Four distinct groups of stakeholders were identified: Non-Governmental Organisations (NGOs), Human Rights researchers, Policy makers and persons with albinism. The first two groups, Non-Governmental Organisations and Human Rights researchers were identified as groups that would have the highest interest in the platform: they are the primary target users of the platform. Policy makers were identified as a group that has less interest in the platform but has the highest decisional power, at different scales. Persons with albinism were the last stakeholder group identified, they have a relatively low interest in the platform but are primarily impacted by the platform, hence why they were accounted for.

The persona created for the first group (NGOs) is Sophie who works for an NGO. Her main goals are tracking accidents and raising awareness around albinism in Africa and what she needs from the platform is to be able to quickly find reliable documents. Sasha is the persona created for the Human Rights researchers group. She is extremely familiar with the topic and needs the platform to find reliable data for specific countries and / or themes. Zehema is the persona created for Policy makers. She is not familiar with the topic and needs to find summarised information quite fast. Finally, Daniel is the persona created for the last stakeholder group, persons with albinism. He will not necessarily use the platform a lot but might find it useful to find other's peoples stories. Figure 2 shows, for each stakeholder persona their backstory, goals, pain points and needs from the platform.



Figure 2: Stakeholder personas

3.2.4 Functional and non-functional requirements

Afterwards, functional and non-functional requirements were defined for the platform. Functional requirements refer to tasks that the platform must perform while non-

functional requirements are related to characteristics of the platform such as speed, security or reliability (Al-Ababneh *et al.*, 2026).

These requirements were defined based on the needs of each stakeholder group. Six functional requirements and two non-functional requirements were identified, these are shown in Table 2.

Functional requirements	Allow the user to search incidents by keywords
	Allow the user to filter incidents by topic
	Show the number of incidents by country on a map
	Show the number of incidents by topic
	Present summary statistics by country
	Show the source of documents
Non-functional requirements	Access to a specific document should be quick
	Documents stored on the platform should come from reliable sources

Table 2: Functional and non-functional requirements

3.3 System design

Once clear requirements were defined, the system could be designed. This step requires designing the architecture of the platform, all the features that it will support and the technology that will be used to build it (Saravanos, 2025). During this step, wireframes of the platform were created.

The platform was designed based on the functional requirements that were defined above. The home page of the platform is shown on Figure 3. It displays general summary statistics: total number of documents analysed, total number of human rights violations found and number of human rights violations for each topic. The user can then click on a “Click for more information” button to access the dashboard.



Figure 3: Wireframe, home page of the platform

The default layout of the dashboard is shown on Figure 4. It has two main components: an interactive map showing how many human rights violations were found in each African country and a search engine, allowing the user to search incidents by keyword and displaying the results below.

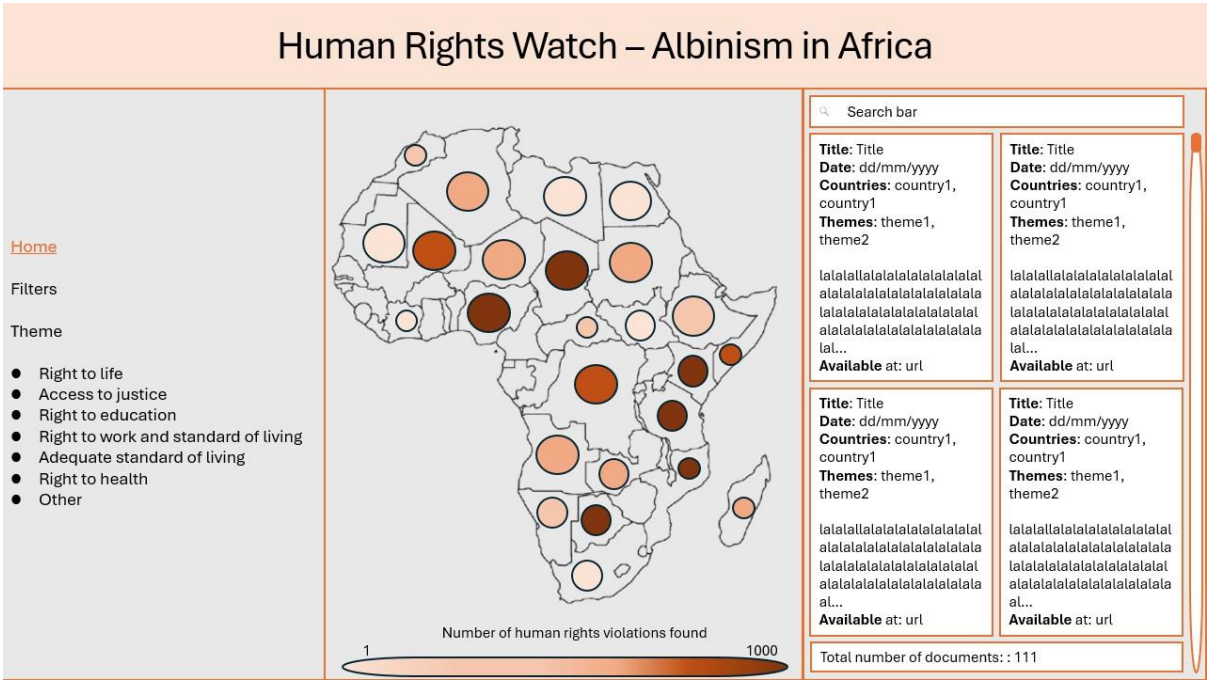


Figure 4: Wireframe, dashboard

As mentioned before, the dashboard allows the user to see only data fitting defined human rights themes. This filter impacts the entire dashboard: the data used to build the map and the documents displayed fit the filter's selection.

The map is interactive: when clicking on a specific country, specific information about the data collected in this country is shown to the user. Each country is coloured in a shade or red; darker shades of red indicate that more human rights violations were found, while lighter shades indicate very few violations were found for this specific country. The documents shown on the right can be searched through. Keywords can be entered in the search bar, reducing the number of extracts displayed to only the ones matching the keywords. Once a keyword has been entered, the data shown on the map is also updated, showing only the data that matches the keyword.

3.4 Implementation phase

The implementation phase is the coding phase, which was made of two stages. The first one was collecting the data that was going to be shown on the platform, and the second one was building the platform.

3.4.1 Data collection pipeline

Based on the wireframes above, the data to be collected was clear. To build this dataset, documents concerning human rights violations of persons with albinism were collected and analysed. Each row of the dataset built was associated with an extract describing a human right violation. For each extract, the URL of the source document, the title of the document, the author or authors of this document, the date of the document, the theme or themes of the extract and the country or countries concerned by the extract were extracted.

The data collection pipeline is presented in Figure 5.

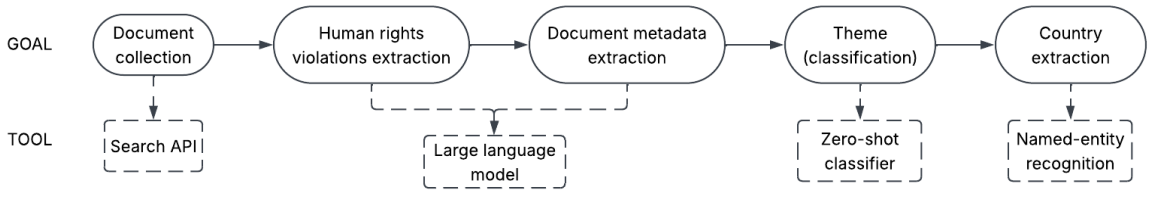


Figure 5: Data collection pipeline

The dataset was constructed through a four-stage process. The first step was to collect documents containing information about human rights violations of persons with albinism in Africa. Then, each document was passed through a Large Language Model (LLM) (Gemini 3.5-Flash model) to retrieve the document's metadata, including the title, author and date and to extract the relevant excerpts from the document. Finally, to complete the dataset, zero-shot

classifier (the MoritzLaurer/deberta-v3-large-zeroshot-v2.0 model) and a named-entity recognition (NER) model (spaCy's en_core_web_sm model) were used to get a theme and a country for each extract.

3.4.2 Building the dataset

Document collection

The instances of human rights violations of persons with albinism were available online in multiple documents. Collecting publicly available documents was therefore the beginning point of the project. A first idea for this would have been to manually search the internet for the extracts, which is a time-consuming task and would not have been feasible, considering the volume of data that was required to have meaningful data to build the platform. Therefore, an automated way of collecting the data was used. The first step was collecting the documents relevant to human rights of persons with albinism, and the second was analysing each document to get details about each human rights violation mentioned in the document.

Several search APIs documentation pages were consulted to select the one that fit the needs of the project the best. The main need of the project was to use a platform that is free to use (or at least has a free plan with a big enough quota). The Exa Search API (*Exa Search API - Exa*) was selected in the final implementation, as it is fast and efficient, but also has a web scraping functionality – it can also retrieve the full content of the website and not only highlights.

To access the most documents possible, multiple different searches were performed using mixes of keywords. Every search contained one keyword from each of the three following categories:

- Mention of albinism using albinism, persons with albinism, children with albinism
- Mention of localisation using Africa, a specific country
- Mention of the topic of the document using human rights violations, killings, discrimination

In order to ensure that no bias was introduced in the dataset by favouring a country, the first searches were made using just the term *Africa*. To complete the dataset, searches were made afterwards using country names. The same search was made with each country name (*Human rights violations of persons with albinism in* completed by each country name).

Excerpts and document metadata extraction

After that, the goal was to extract the parts clearly mentioning human rights violations of persons with albinism from the documents. Because the search API's user's conditions do not

allow the user to keep and store the full results returned by the API, the text was analysed immediately after the URL collection, and the full content of each website was discarded.

Each full document was passed through the Gemini API (*Gemini API*, 2026) to identify and collect the human rights violations described in each document. The Gemini-3.5-Flash model was used. Additionally, for each document, the title, date and author of the document were collected if they were available.

The following prompt was used: *Return the detailed instances of human rights violation against a specific person with albinism that happened in Africa*, followed by the text of the document currently being analysed. The response returned by the model was carefully structured to ensure that it could be used automatically without treatment. The model was asked to return five items:

- n_violations: an integer, the number of human rights violations found in the document
- violations_sentences: a list containing the exact sentences that contain each of the human rights violations mentions
- title: an optional field, the title of the document being analysed
- author: an optional field, the author of the document being analysed
- date: an optional field, the date of the document being analysed

Theme classification

Extracting a theme from each extract was done using zero-shot classification (mentioned in the literature review). Zero-shot classification was chosen for this task because it did not require any labelled data. No labelled data was available for this specific classification task: it would have had to be specifically created for this, which would have been quite time-consuming considering the time that was allocated to this project. The performances of the zero-shot classification model were good enough for this purpose.

There are multiple zero-shot classification models available. Multiple of them were tested on a few test extracts. Finally, the model used was the DeBERTaV3-Large (He *et al.*, 2021), which was mentioned in the literature review. It was chosen because it was the model that worked the best on the test extract, and in a reasonable time for the number of lines to analyse (around thirty seconds per extract).

Country extraction

The second part of the extract analysis was extracting the country mentioned in the extract. After searching for ways to do this, it was decided to use named-entity recognition. Two different approaches were tested for this task. The first one was using a model that recognised

both the places' names and the country associated with them. This approach was the simplest one to implement, hence why it was used in the first place. However, after analysing a few of the results produced by this model, it turned out that it would mark quite a lot of extracts as mentioning the United States or European countries with no apparent reason other than generic names being more easily attributed to these countries.

Because of that, a second approach was used: using two different models, one for the named-entity recognition – a model that extracts the names of places mentioned in the extract, and the other to map the place name to a country. This approach gave better results, even though a lot of extracts were still attributed to wrong countries, which was manually fixed.

3.4.3 Building the platform

Once all the data was collected, the platform could be built. The main two features of the platform were an interactive map of Africa to visualise the data collected by country and a search bar and the associated results. Therefore, the tool that was used to build the platform had to support both these features. Streamlit (Streamlit, 2021) was the first platform building tool that was tested, because of how easy to use it is. As these two features were easily and well-integrated using this tool, it was selected for the final implementation.

3.5 Testing

The testing phase was the final important phase during which the platform that had been built got tested in order to make sure it met the requirements that were defined (Saravanos, 2025).

The platform evaluation process required primary data collection. This step was crucial for the project. Having feedback on the platform given by persons outside of the project is important as it gives insight on how the platform is perceived by others. Ethics approval was obtained, and relevant ethics documents are presented in appendix D.

To collect the data, an anonymous questionnaire was used, followed by a quick exit interview. The questionnaire had two distinct parts. The first part was based on the functional requirements that were defined for the platform in the platform design part. This part asked the respondents to perform different tasks on the platform before evaluating the ease with which they performed the task on a scale of one to ten. The second part was based on the System Useability Scale (SUS) (Sauro, 2011). This scale is a way of obtaining feedback on the general user experience on the platform. It contains ten statements about the user's experience and asks the user how much they agree with each statement, on a scale of one to five.

The table below show each of the questions that were incorporated in the questionnaire, and the reply format.

Part	Question	Answer format
First part	Evaluate the feasibility of the task: get access to the number of incidents that were found	Number between 1 and 10
	Evaluate the feasibility of the task: search an incident by keyword	
	Evaluate the feasibility of the task: filter the incidents by topic	
	Evaluate the feasibility of the task: get the number of incidents for a specific country	
	Evaluate the feasibility of the task: access a document based on an extract	
Second part	Do you agree with the statement: “I think I would like to use this system frequently”?	Number between 1 and 5
	Do you agree with the statement: “I found the system unnecessarily complex”?	
	Do you agree with the statement: “I found the system easy to use”?	
	Do you agree with the statement: “I think that I would need the support of a technical person to be able to use the system”?	
	Do you agree with the statement: “I found the various functions in this system were well integrated”?	
	Do you agree with the statement: “I found there was too much inconsistency in this system”?	
	Do you agree with the statement: “I would imagine that most people would learn to use this system very quickly”?	
	Do you agree with the statement: “I found the system very cumbersome to use”?	
	Do you agree with the statement: “I felt very confident using the system”?	
	Do you agree with the statement: “I needed to learn a lot of things before I could get going with this system”?	

Table 3: Questionnaire details

The replies to the second part of the questionnaire were used to compute a SUS score for each user. A normalisation step was performed first: the replies to odd items (positive statements) had 1 subtracted to them and the replies to even items (negative statements) were subtracted to 5, before summing all the answers. Finally, the score obtained was multiplied by 2.5, to obtain a score between 0 and 100 (Sauro, 2011).

Exit interviews were then conducted with each of the respondents, either in person or through the phone, to identify clearly what worked well and what did not. The first questions were questions about the respondent's general impression of the platform. The next questions asked varied depending on the answers to the questionnaire. In places in the first part of the questionnaire where low numbers were selected, follow-up questions were asked during exit interviews to identify exactly what did not work and how it could be improved.

Finally, the improvement areas that were identified during exit interviews were used to improve the platform.

3.6 Implementation details

The project was implemented with Python 3.12.4, using VScode. Microsoft excel v2607 was used to quickly visualise the data and for the analysis of the questionnaire responses. During the data collection process, the following python packages were used:

- pandas (version 3.0.3)
- exa_py (version 2.16.0)
- google-genai (version 2.12.1)
- pydantic (version 3.7.4.3)
- typing (version 3.7.4.3)
- torch (version 2.7.1)
- transformers (version 4.52.4)
- spacy (version 3.8.14)
- geopy (version 2.5.0)

During the platform building process, the following python packages were used:

- streamlit (version 1.60.0)
- pandas (version 3.0.3)
- numpy (version 2.5.1)
- plotly (version 6.9.0)
- bm25s (version 0.3.10)

CHAPTER 4 RESULTS AND ANALYSIS

This section details the results that were obtained throughout the project. The section is organised in four parts. The first part focuses on the dataset that was built throughout the project. The second part details the various features that have been implemented, using the dataset built previously. The third part focuses on the testing phase of the platform, using the questionnaire and exit interview responses. Finally, the fourth part shows the final version of the platform.

4.1 Building the dataset

4.1.1 Document collection and excerpts extraction

The first step of the dataset building, as mentioned in the methodology section was to analyse documents mentioning human rights violations of persons with albinism. The main python function used to collect the data is available in Appendix A.1.

4.1.1.1 Initial data collection

159 distinct documents were analysed in total during the document collection process. 94 human rights violations against persons with albinism in Africa were found in these 159 documents. On average, 5 human rights violations were found in one document.

4.1.1.2 Cleaning the dataset

The data collected was cleaned. This step was crucial to ensure the results were as reliable as possible. There are still probably errors in the collected data, as there was no time to look through every row of the dataset, but the data cleaning phase drastically reduced the number of errors, thus making the end results reliable.

Duplicated documents

The first step of the data cleaning step was to look for duplicates. Even though one URL was never analysed more than once, the same document could be available at two different URLs. Because the excerpts were extracted using a large language model, the results of passing the same document twice in the same model could differ. To identify documents that had been analysed twice, the extracts column of the dataset was analysed to find duplicates. 38 extracts were present more than once in the dataset, corresponding to 10 documents. For each of the duplicated extracts, all extracts coming from the corresponding URLs were analysed. In most cases, the extracts that came from a document that had been analysed for a second time roughly matched the extracts that came from the first analysis of the document. There were only small differences in most of them (the Gemini model would include one more point or word). Therefore, it was decided to remove from the dataset all extracts from the second time a document had been analysed.

A second step was performed to identify duplicated documents that would not have the same extract twice. The title column of the dataset was analysed to look for duplicated documents. There were four documents that fit this description. Again, because the excerpts extracted were similar between the two analyses of each document, the ones obtained from the second analysis were discarded.

After this step, the dataset contained 558 rows coming from 145 distinct documents.

Date and author cleaning

The next step was to clean the columns containing the author and the date. A first analysis was conducted by looking through the dataset manually and inspecting the date and author columns. The authors and dates that were collected were found to be reliable.

The main task when cleaned these two columns was filling the rows that were either missing an author or a date. 32 documents out of the 145 that were analysed were missing an author, meaning that 22% of the documents were missing and author. These documents were looked at, and their author or authors were manually added to the dataset when an author was available.

For 13 out of the 32 documents missing an author, an author was found and added to the dataset (40.6% of the documents missing an author). For 18 of these documents, no author was mentioned on the website (56.3% of the documents missing an author). Finally, it was found that the last document was not relevant to human rights violations of persons with albinism and was therefore removed from the dataset.

49 documents out of the 145 that were analysed were missing a date, representing 32.8% of the documents. A date was added for 37, or 75.5% of the documents. The other 12 documents (24.5% of the documents) had no date mentioned on their website.

Once the cleaning was finished on the author and date columns, 12.5% of the 144 documents analysed were missing an author and 8.3% of these documents were missing a date.

The final cleaned dataset contained 556 human rights violations against persons with albinism coming from 144 distinct documents.

4.1.1.3 Overview of the collected data

This section provides an overview of the data that was collected.

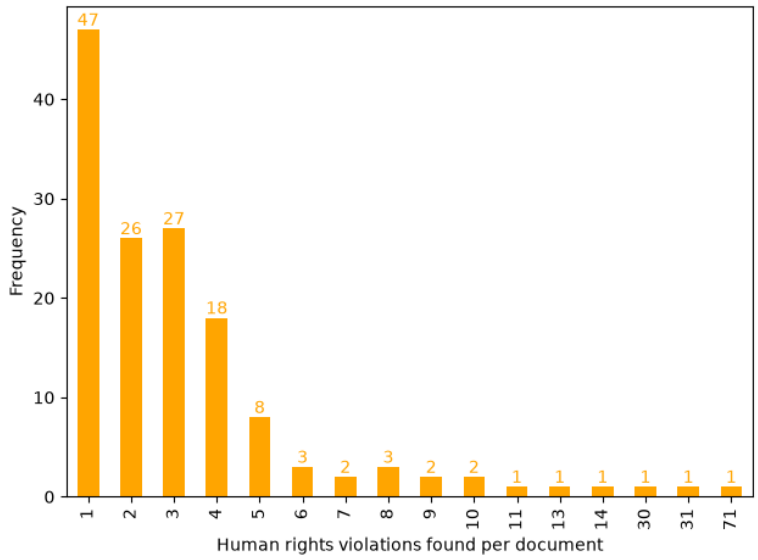


Figure 6: Frequency of the number of human rights violations per document

Figure 6 above shows the frequency of the number of human rights violations per document. 1 human right violation was extracted from many documents (47 documents out of 144). Most documents contained less than five human rights violations which were extracted. 3 documents contained an unusual number of extracted human rights violations: respectively 30, 31 and 71. To ensure that these high numbers were not hiding errors, the extracts coming from each of these 3 documents were analysed carefully. No irregularity was found in the dataset. The explanation to these high numbers is that these documents are very long and detailed, and contain many information.

Examples of excerpts can be found in the table below. The lengths of the excerpts extracted from the documents vary between 36 ("*the 2015 murder of Elizabeth Kwagala"*, 6 words) and 1224 characters (around 230 words). The mean extract length is just under 200 characters (196 characters) while the median is 177 characters.

URL	**Extract**
https://africaalbinismnetwork.org/wp-content/uploads/2025/04/access-justice-person-with-albinism.pdf	*"According to the official court record shared by albinism groups, on 23 January Kasoma Isaac and Casius Chiyunda kidnapped MC, a 13 year-old boy with albinism, and his friend who happened to be with him at the time. (Zambia)"*
https://www.african-court.org/ajudata/download-file/decision/689/EN/179	*"the education system provided at the shelters did not cater to the rights of PWA as for example the children with albinism were not provided with assistive devices such as*

	spectacles. Consequently, the Court found a violation of the right to education."
https://www.underthesamesun.com/wp-content/uploads/2026/01/Attacks-of-PWA-Extended-23.01.2026.pdf	*"On 18th April 2025, a boy with albinism by the name of F.H. was abducted in Mombe Village, Mulanje district, Malawi, though later rescued."*
https://africaalbinismnetwork.org/wp-content/uploads/2024/07/Alternative-Report-on-PWAs-in-Ghana-to-the-CRPD-by-AAN.pdf	*"When I was born, my father rejected me. My mother told me that he gave me poison to drink, but I vomited it out and survived. I was then forced to serve a fetish priest, whom I had to accompany and serve anytime they were in a trance."*
https://www.knchr.org /Portals/0/Albinism%20Mapping.pdf	*"Another respondent also mentioned how her neighbours had on two occasions tried to evict her from her property through acts of arson. In both these incidents, the relevant authorities were notified but no substantive action was taken against the perpetrators."*

Table 4: Examples of extracts

Table 4 shows five example of extracts and the different documents from which each of them have been extracted.

4.1.2 Extracts analysis

Each extract was then analysed. The main python function used for analysis is available in Appendix A.2.

4.1.2.1 Theme classification

After being collected, each extract was passed through the zero-shot classification model to assign it one or multiple themes. As mentioned in the methodology section, the possible themes were right to life, right to education, right to work, access to justice, adequate standards of living and right to health.

The model used as inputs an extract, a hypothesis and the available themes. The hypothesis is a statement that the model adds before each theme to construct a full sentence, which is then compared to the extract, as explained in the literature review section. The model then attributes a score to each possible theme and returns these scores. A threshold score was then chosen to determine which themes were assigned to the analysed extracts.

A threshold of 0.4 was chosen, meaning that a theme that had a score higher than 0.4 for a given extract would be assigned to this extract. This threshold was chosen after passing

multiple test extracts through the classifier and choosing a threshold that would classify assign at least a theme to enough extract, without assigning extracts to non-sensical themes.

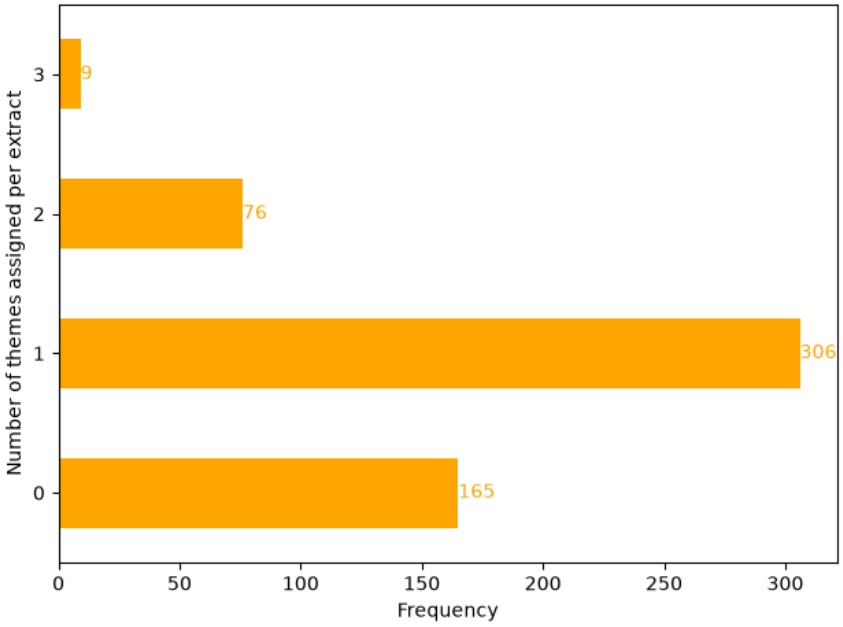


Figure 7: Repartition of the number of themes per extract

Figure 7 shows the repartition of the number of themes assigned by country. Out of the 556 rows in the dataset, 165 were not assigned any theme (29.7% of the total number of extracts), 306 were assigned exactly one theme (55% of the total number of extracts), 76 were assigned exactly two themes and the last 9 were assigned three themes.

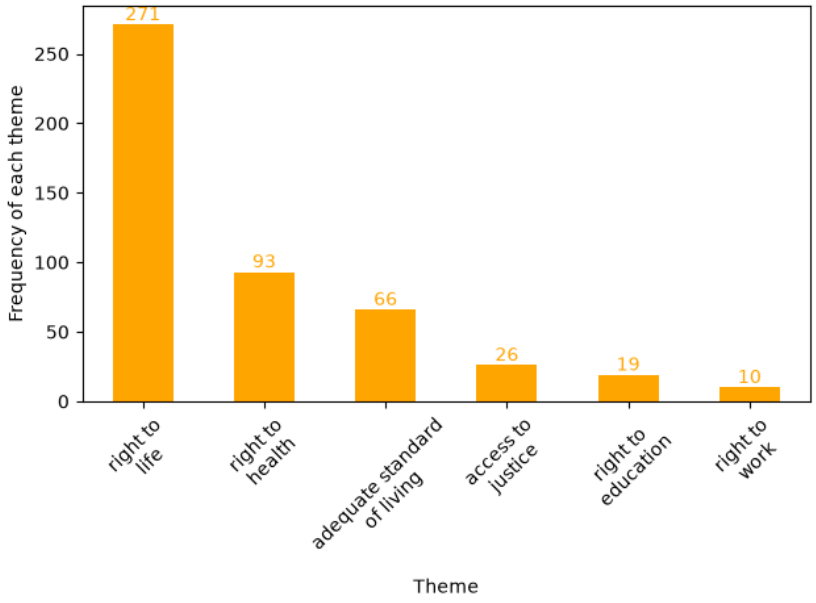


Figure 8: Number of extracts fitting each theme

Figure 8 shows the number of extracts that were assigned each theme. A majority of extracts were assigned the theme *Right to life* (271), which might seem like a lot. This high number can be explained by the fact that killings or abductions constitute the most serious cases of human rights violations. These cases are therefore more often reported to the police, and afterwards in the media. Consequently, it makes sense that most of the human rights violations that were extracted from various documents mention violations of the right to life of persons with albinism.

The two least represented themes in the dataset are *Right to work* and *Right to education* with respectively 10 and 19 human rights violations related to these themes found in the dataset. The theme *Access to justice* is also sparsely represented in the dataset, with 26 extracts having been assigned this theme. Finally, 66 extracts have been assigned *Adequate standards of living* and 93 have been assigned *Right to health*.

Table 5 shows, for each theme, an example of an extract which has been assigned this theme and an example of an extract which has not been assigned any theme.

URL	**Extract**	**Theme**
https://www.ungeneva.org/en/news-media/meeting-summary/2022/08/experts-committee-elimination-racial-discrimination-commend	*"There was also a case of a young girl with albinism who was abducted and disappeared who had been found by the police."*	Right to life
https://africaalbinismnetwork.org/wp-content/uploads/2024/09/Alternative-Report-on-Benin-to-CEDAW.pdf	*"For example, a young lady with albinism was initially denied a teaching position at a school, with the justification that his visual impairment was a barrier, an excuse that was clearly a pretext for discrimination. It was only after intervention by an albinism advocacy group that he was hired, and he later received commendations for his exceptional performance."*	Right to work
https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0205774	*"During my primary [school] we had a teacher who used to chase me out of class whenever she entered saying "you're not supposed to be here"."*	Right to education

https://www.iwacu-burundi.org/englishnews /burundi-albinosdiscrimination -a-sad-situation/	*"According to Nathalie Muco, Chairperson of "Albinos Women Association Hope", albinos are not accepted in their community. They are vulnerable, lonely, unemployed, assaulted, kidnapped or killed, weak and their security leaves to desire."*	Adequate standard of living
https://www.institut-fuer-menschenrechte.de/fileadmin/Redaktion/ PDF/DB_ius/CRPD_18082017_ X_v_Tansania_ENG.pdf	*"The Committee also considers that the suffering experienced by the author owing to the lack of action by the State party that would allow the effective prosecution of the suspected authors of the crime, becomes a cause of re victimization, and amounts to psychological torture and/or ill-treatment."*	Access to justice
https://collaborativemedia.wscij.org/nigeria-albino-communitys-many-woes-denial-of-voting-education-health-and-financial-inclusion-rights/	*"Another advocate of persons with albinism cried against injustice in the health rights of the Albinism community. He maintained, a major challenge of the Albinism community is their health, which is related to their skin textures. He said "Many persons with Albinism had died prematurely because of skin cancer"."*	Right to health
https://digitalcommons.wcl. american.edu/cgi/viewcontent. cgi?article=2086&context=hrbrief	*"In some instances, even the graves of deceased PWA have been desecrated for their remains."*	No theme found

Table 5: Examples of extract for each theme

4.1.2.2 Country extraction

Each extract was also assigned a country, using the methodology that has been discussed in the previous section. Once the extraction was completed, 195 extracts out of the 556 rows in the dataset had no country identified. However, as mentioned in the methodology section, non-African countries were assigned to many extracts. Therefore, the first step in the cleaning process of this column was removing the non-African countries from the columns. Once this was done, 42 more extracts had no African country identified, meaning that a total of 237 rows had no country identified (42.6% of the rows).

Because the proportion of rows missing a country was quite high, another country extraction method was taken. All extracts that had no country were analysed to determine whether an African country had been mentioned in the extract, the title or the URL of the document but had not been detected by the Named Entity Recognition model. These omissions were caused by the fact the NER model only recognises places as places when they are explicitly mentioned as such in an extract. For example, the model might not recognise “Mali” as a place in the sentence “Mali: a man was found dead”. Therefore, a list of African countries was created based on the ones that had already been recognised, and string matching was applied to all extract, titles and URLs having no countries mentioned. This was a very effective step, as the number of extracts with no country identified went down from 237 to 136, meaning that 99 countries were identified through this process.

Finally, the remaining 136 rows missing a country were manually investigated. This was mainly done to ensure that all documents collected were about African countries, but to perform a final check that the attributes of the other columns – date, title, author and themes - were correct. Once this was done, 63 rows remained without any country identified, but there was no doubt that these rows still concerned African countries. Some extracts only mention “Africa” or “Sub-Saharan Africa” without a specific country.

The last check that was performed on the country column was checking that countries which could be mistaken, such as the Democratic Republic of Congo (DRC) and the Republic of Congo (RC) or Guinea, Guinea Bissau and Equatorial Guinea. Because no extract had been identified as mentioning the Republic of Congo, the extracts that had been identified as mentioning the DRC were manually analysed. One row was misidentified and assigned DRC even though it mentioned Brazzaville, the capital city of the RC, proving that the two countries were not necessarily distinguished. This mistake was manually fixed. This was the last step of the data cleaning process for the country feature of the dataset.



Figure 9: Countries mentioned in the dataset

The figure above shows which African countries were identified and which countries were not. This was particularly interesting to ensure the data was as complete as possible. The map shows that most of the countries which are not represented in the dataset are in North Africa, in countries such as Algeria, Tunisia or Libya. This is not a cause for concern as multiple reports mention that persons with albinism tend to face more discrimination in countries where they are particularly visible. This would explain the lack of reports of human rights violations in North African countries. Apart from Equatorial Guinea and islandic countries (Comoros, São Tomé and Príncipe and the Seychelles), all Sub-Saharan countries have been included in the dataset, there is at least one extract related to each Sub-Sahara country).

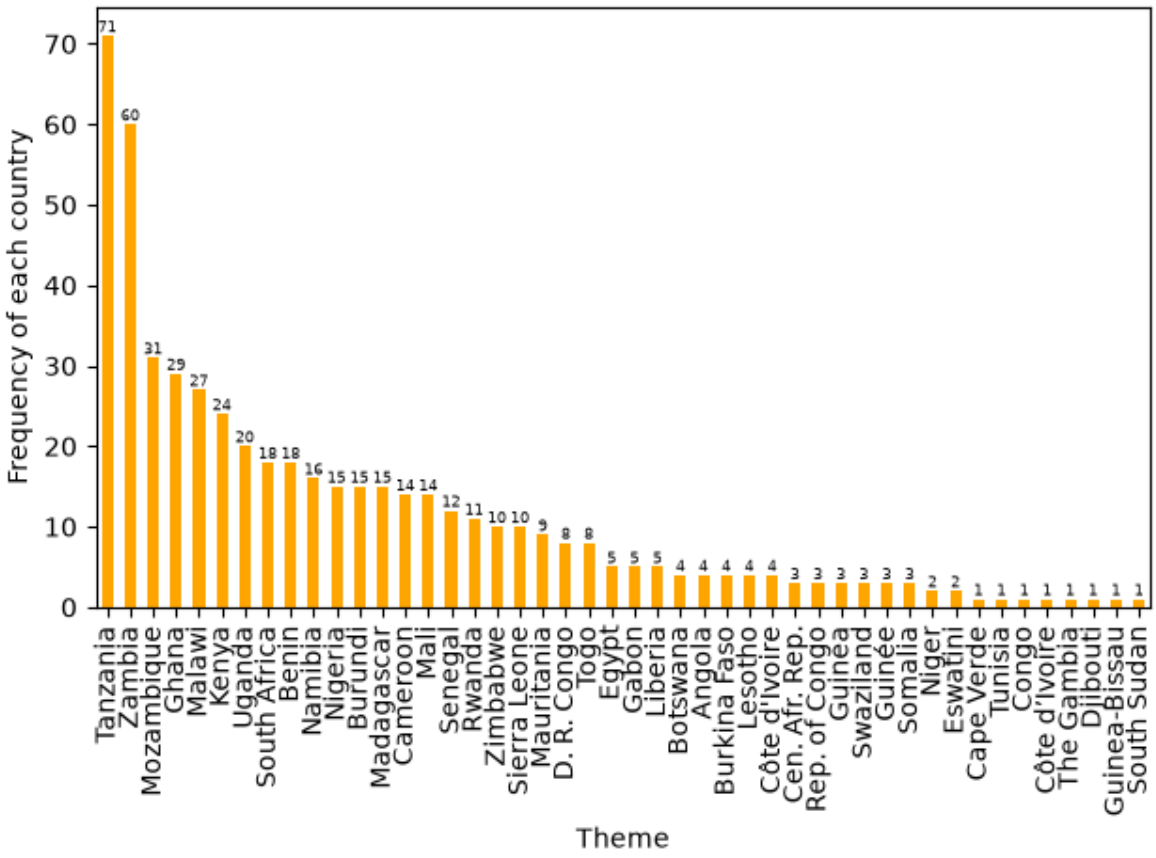


Figure 10: Number of human rights violation for each country

Figure 10 above shows the number of human rights violations for each country. The country with the most reported human rights violations is Tanzania, with 71 incidents found. Countries neighbouring Tanzania also have a high number of reported incidents: 60 were found in Zambia, 24 in Kenya, 31 in Mozambique and 27 in Malawi. This is coherent with the various reports that were analysed before, which mentioned that Tanzania was an epicentre of violences cases against persons with albinism.

4.1.2.3 Excerpts in foreign languages

Some documents that were collected were not in written in English. This is because many of the documents collected come from local newspaper of specific countries, and these newspapers do not necessarily publish articles in English. Documents from Angola or Guinea Bissau were in Portuguese and documents from Senegal or Ivory Coast were in French. A specific attention was given to these extracts, as Natural Language Processing models can underperform on text that is not in English. All the models that were selected have mostly been trained on English text, and even though they can generalise to other languages, the performances tend to be slightly worse when used on text not in English (Renjit and Idicula, 2024).

To ensure that extracts not in English also had accurate themes and country, they were manually investigated. Extracts in French were investigated first. As I speak and understand French, no translation software was needed. A few extracts in French were analysed, and it was concluded that, when they had been available, the themes and countries that were extracted matched each of the extracts. It was then induced that extracts in Portuguese were also correctly classified.

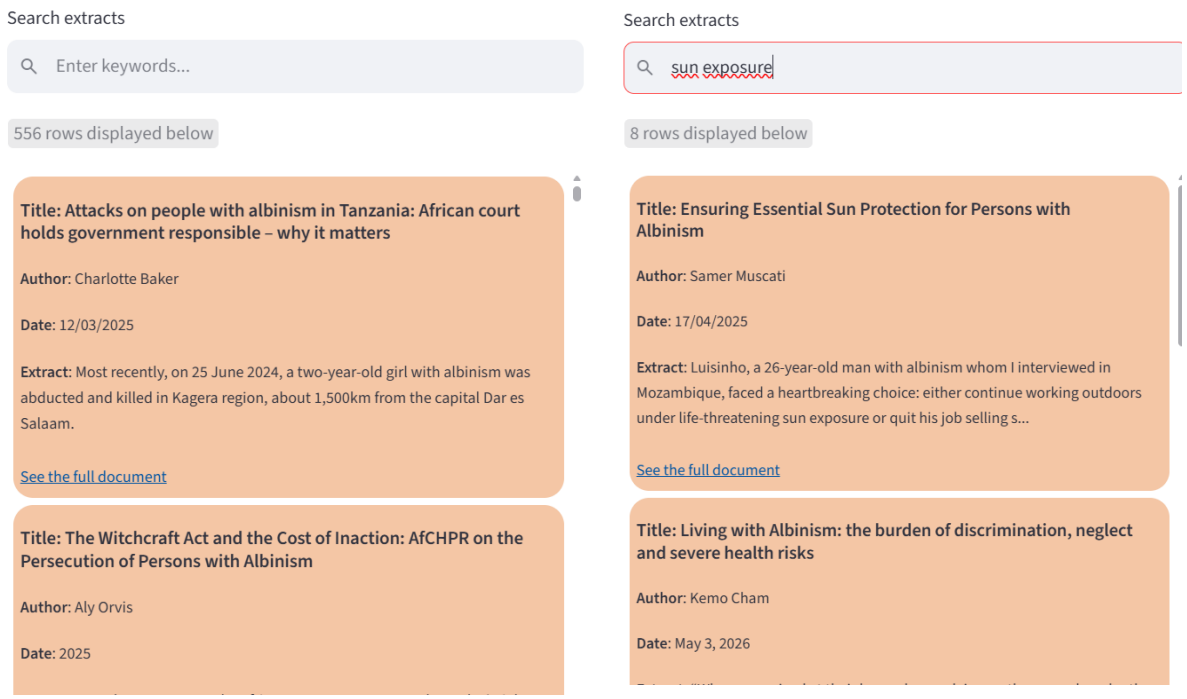
31 extracts in the dataset were in French and 4 in Portuguese. Out of the 31 extracts in French, 16 had not theme assigned to them, meaning that 52% of the extract were not assigned a theme. This proportion was way higher than the overall proportion of extract that were not assigned any theme. However, the small number of samples meant that this could not be a true cause of concern. The 4 extracts in Portuguese were all assigned a theme. Considering the models chosen could still perform relatively well on the extracts in other languages and the small sample size of extracts in foreign languages (35 extracts, 6.3% of the total number of extracts), it was not deemed necessary to use other models trained on French or Portuguese data on the French and Portuguese extracts.

4.2 Building the platform

Once the dataset had been built, the next step was to build the platform. The section details how the two main components of the platform function: the search engine and the interactive map.

4.2.1 Search engine building

The first important part of the platform that was built was the search engine. On the platform, this is materialised as a search bar and a scrollable list of results below. By default, all documents are displayed on the list. The figure 11.1 shows the default version that is shown to the user upon arriving on the platform.



11.1: Default search engine

11.2: Used search engine

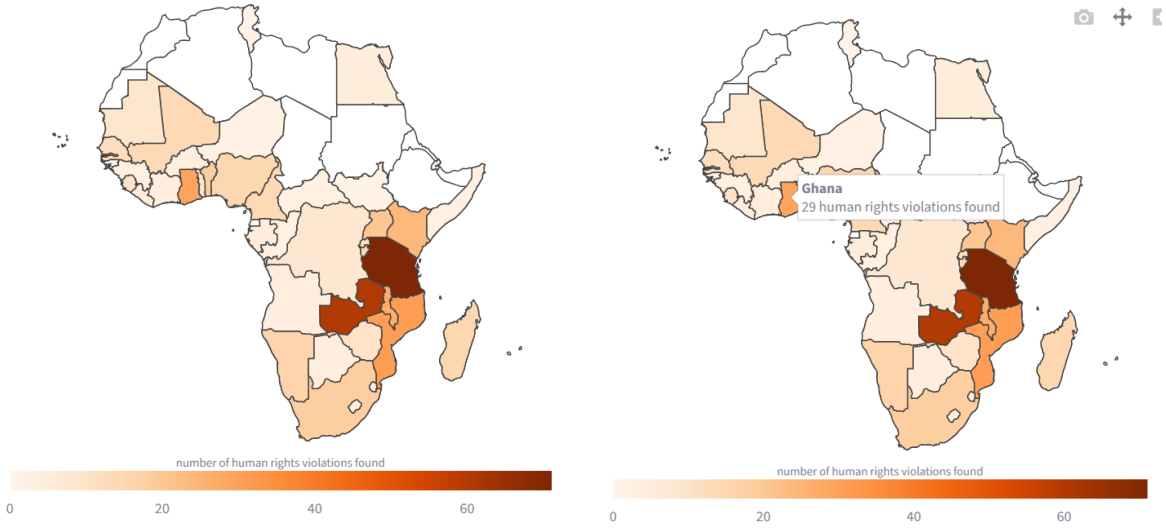
Figure 11: Search engine

To visualise only documents fitting a certain theme, the user can then enter keywords in the search bar, which then updates the list of documents shown below. The bm25 algorithm, mentioned in the literature review, is used to rank all extracts in the dataset. All extracts are given a score and ranked based on their score. Due to the structure of the algorithm, the scores cannot be interpreted by themselves. Therefore, a unique threshold to decide which documents to keep could not be used. The solution was defining the threshold as half the score of the highest scoring document. Every document that has a higher score than this threshold is then displayed on the scrollable list. Documents with a higher score are displayed first. The figure 11.2 shows the result seen by the user when typing *Sun exposure* in the search bar.

4.2.2 Map implementation

The second main component of the platform is the interactive map. It summarises the data that was collected for the user in a quick and efficient manner. The figure 12.1 shows the default version of the map.

To add interactivity and information, when the user hovers over one country, details about the data collected on this country are shown. The figure 12.2 shows this behaviour, when hovering over Ghana.



12. 1: Default map

2: Hovering function

Figure 12: Interactive map

4.2.3 First version of the platform

The home page of the first version of the platform is shown on Figure 13. This page is meant to quickly show useful, aggregated data on human rights violations against persons with albinism. The number of human rights violations per theme is shown on this page.



Figure 13: Home page of the platform

Figure 14 shows the first version of the dashboard of the platform.

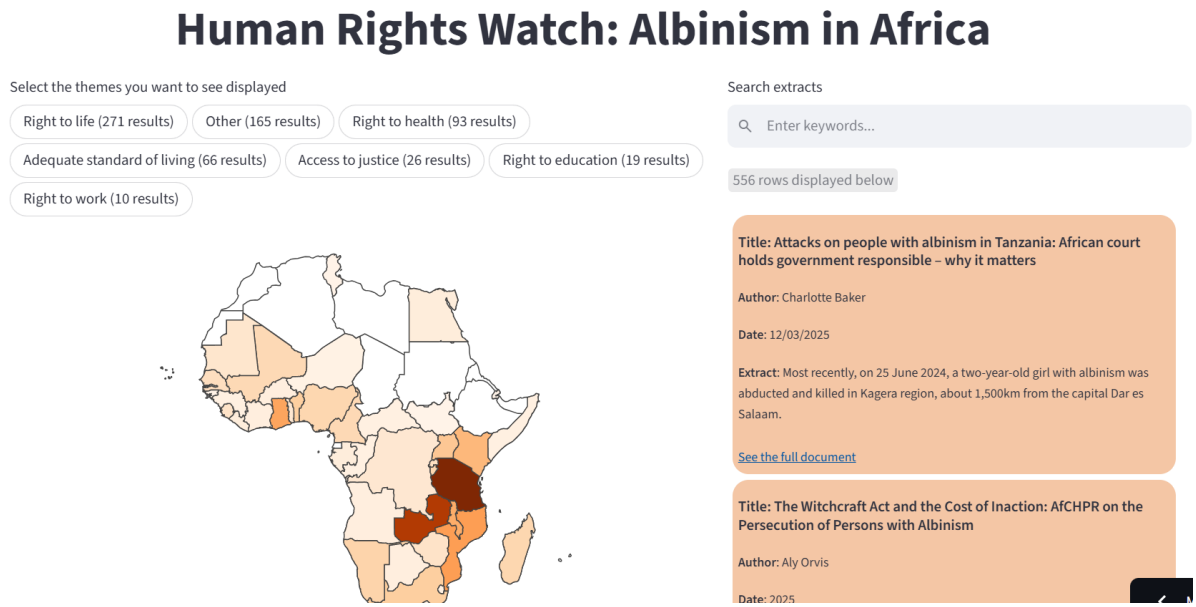


Figure 14: Dashboard

On the left, the interactive map that was mentioned before is displayed. The search bar and the results associated are shown on the right of the dashboard. The dashboard also allows the user to filter the incidents based on the themes associated with them. The map and the search results displayed are updated when one or more themes are selected.

4.3 Testing phase

Once a first satisfactory version of the platform was obtained, it was tested using the process that was described in the methodology section. 12 responses to the questionnaire were recorded. All responses were complete and were therefore used for the analyses below.

4.3.1 SUS questionnaire

SUS scores were calculated based on the feedback that was given, following the methodology that was described in the section below. The figure below shows the repartition of the SUS scores.

The average SUS score for the platform was 92 which meant the platform functioned generally really well. Figure 15 shows the repartition of SUS scores across respondents. Overall, the SUS scores ranked between 82.5 and 100, which indicated that, for every respondent, the platform performed very well.

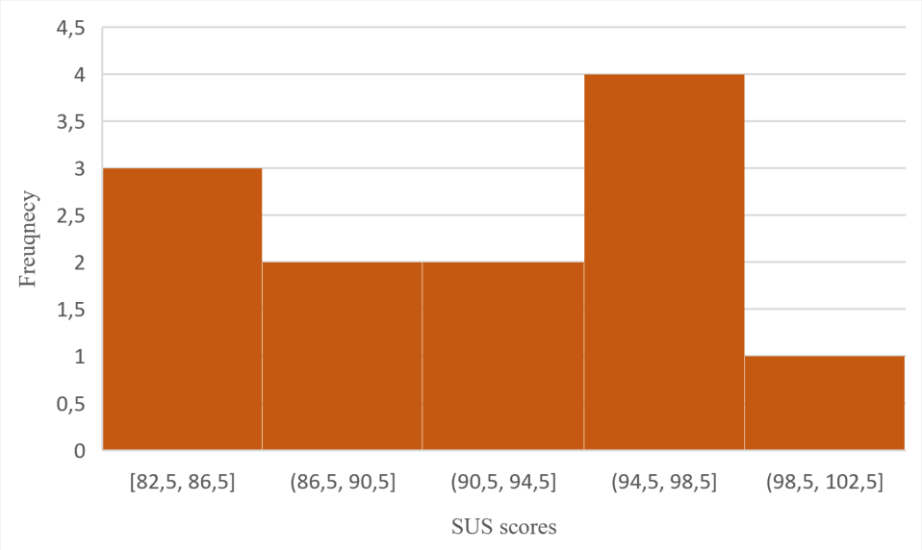


Figure 15: Repartition of SUS scores across respondents

4.3.2 Task-specific questions

Respondents were asked to perform five different tasks on the platform before rating them on a scale of 1 to 10. To make this report understandable, tasks were numbered from 1 to 5, based on the order they appeared in the questionnaire. All five tasks were performed easily according to the respondents' answers. The average answers for every task were all around 9. Shown below, on Figure 16, are the distributions of the answers that were given for each task. Task 1 (Getting access to the number of incidents), task 3 (Filtering the incidents by topic) and task 5 (Accessing a document based on an extract) all had ratings of 8 and above for every respondent. They were straightforward and easy to perform tasks to perform on the platform.

Task 2 (Searching an incident by keyword) and task 4 (Getting the number of incidents per country) also had overall great ratings (7 and above for every respondent), indicating that respondents managed to perform the task pretty well. They were, however, the lowest rated tasks. During the exit interviews, questions on possible improvements for both these tasks were asked.

For task 2 (Searching an incident by keyword), the improvement that was proposed was adding shortcuts to popular keywords. This was difficult to implement, but further questioning revealed that providing the user with examples of keywords to type to the search bar was the main idea behind the request. Examples of keywords that could be written were added on the default text shown to the user on the search bar, which was modified

For task 4 (Getting the number of incident per country), the improvement that was mentioned by multiple respondents was adding text on the platform to explain to the user that they could hover on countries. This was easily added to the platform to address the issue.

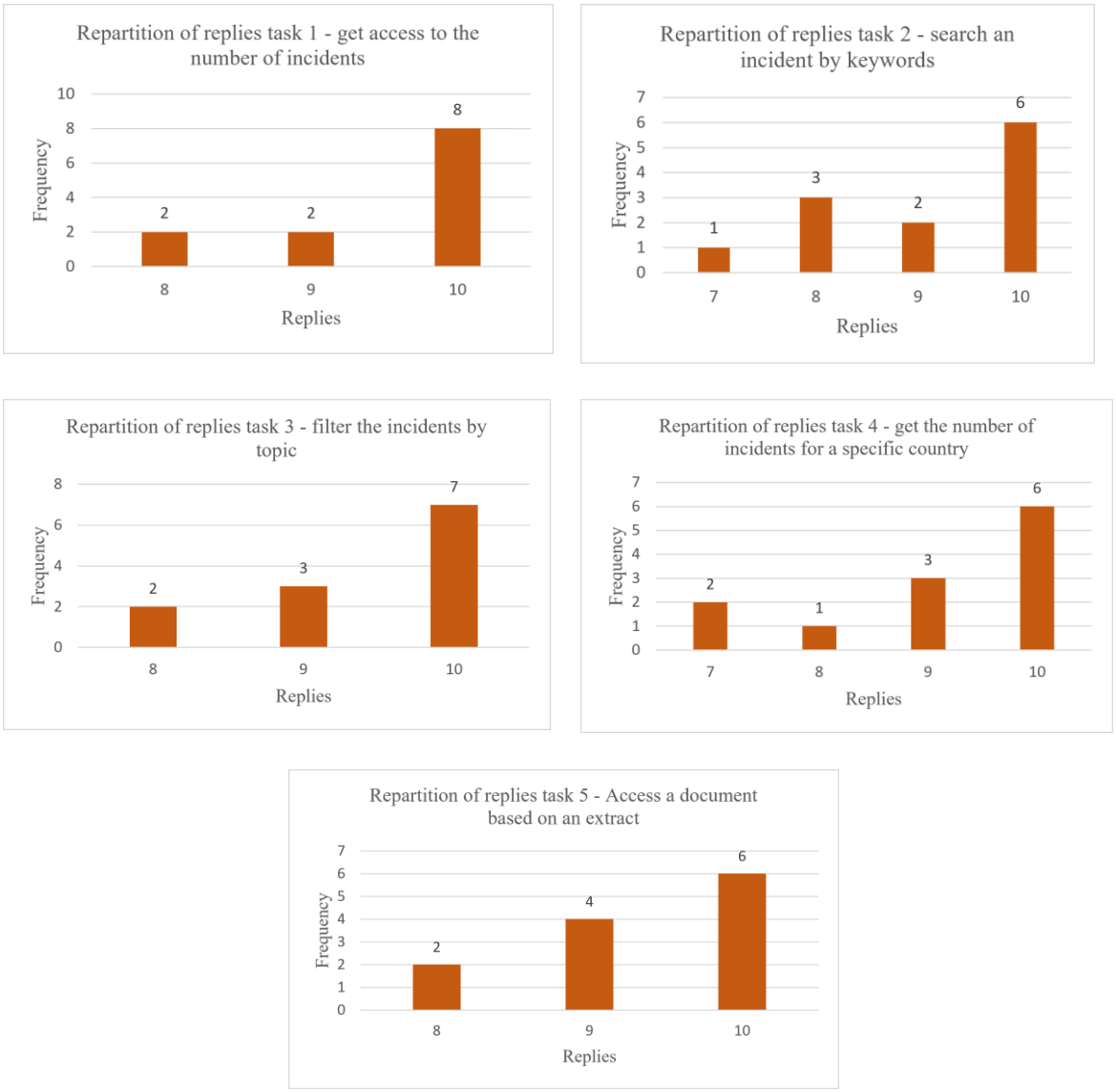


Figure 16: Distribution of replies for each task

Other issues were brought up during the exit interviews. The first one was that the “see more” button on the Home page was not extremely visible. One respondent mentioned that, at first, they did not understand that they had to click on the button to access the dashboard. The background colour of the button was changed from white to orange to make it more visible on the page.

Another issue that was brought up multiple times was that the contrast between the colour of the text and the colour of the boxes was very light when using the platform on dark mode. Because the platform was not built to be visualised in dark mode, to prevent the user from

using the platform on dark mode, the default mode of opening of the platform was switched from automatic to light. This way, the platform would never be opened in dark mode. Further improvements can be made by changing the entire colour scheme of the platform when in dark mode. As this was a very minor issue, this was not implemented.

4.4 Final platform

Once the necessary adjustments had been made, the final platform was deployed. The final version of the home page, along with the modifications that were mentioned in the previous part, is shown on the figure 17.

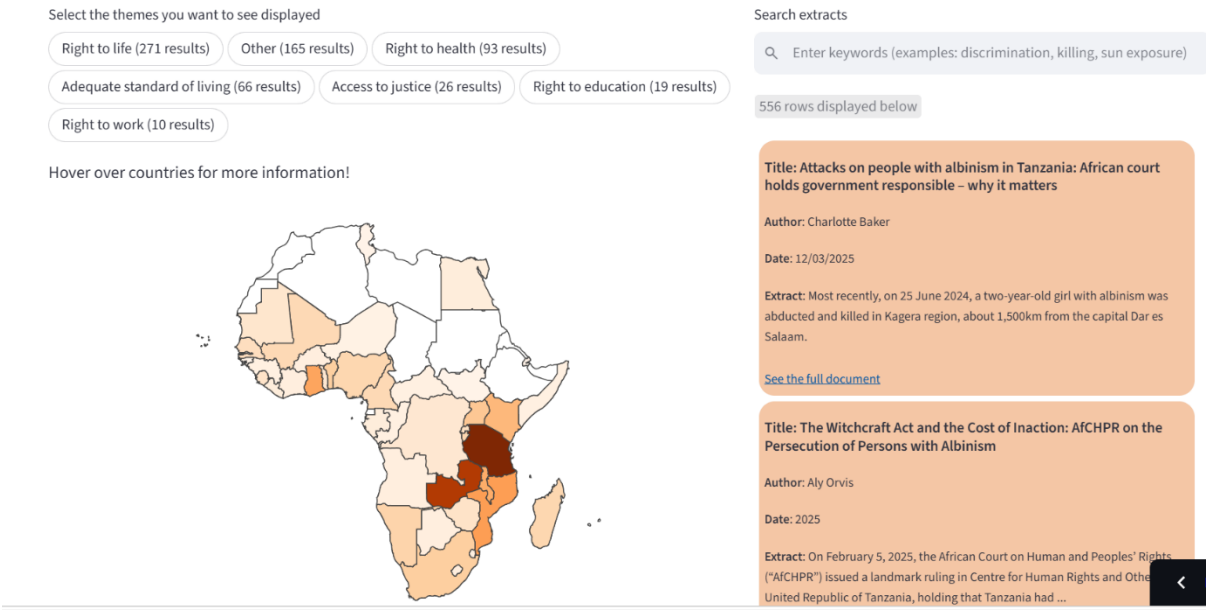


Figure 17: Default dashboard, final version

Figure 18 shows the dashboard when a theme has been selected. The theme *Right to health* was selected. Consequently, the colours on the map were updated to match the new data and the data that is shown on the scrollable list is the data that has been assigned the theme *Right to health*.

The platform is available at the following link: Home Page · Streamlit

The code used to build the platform is available in Appendix B.

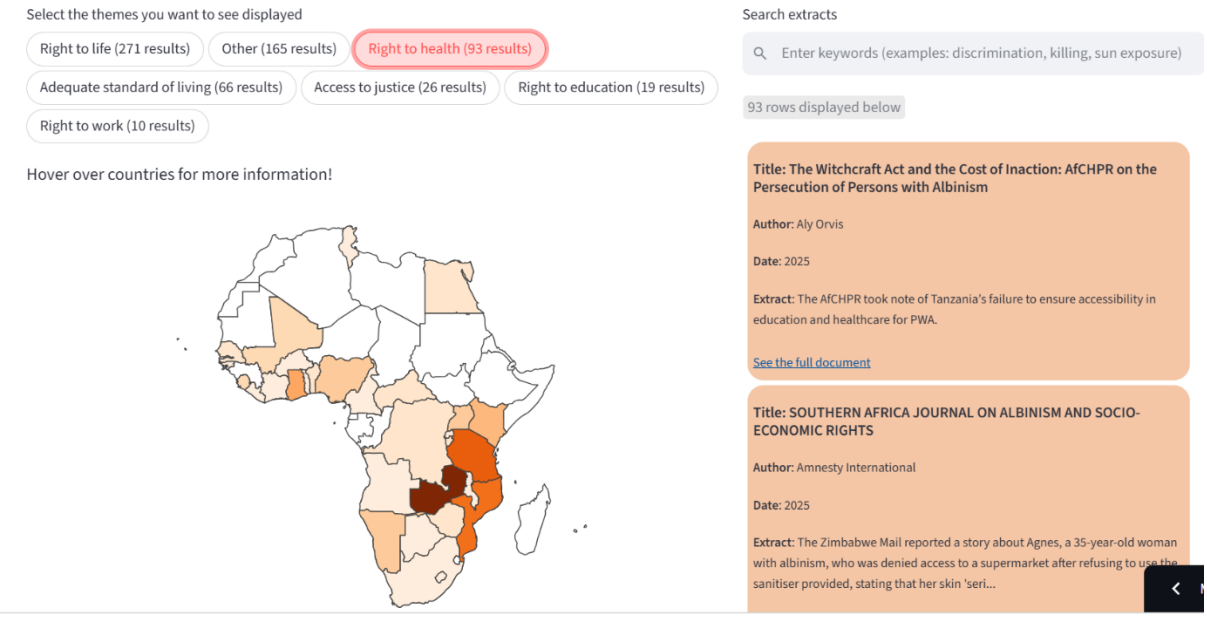


Figure 18: Dashboard with selected theme, final version

More wireframes of the final version of the platform are available in Appendices C. The user can select multiple themes to see displayed. The result of selecting both *Right to work* and *Right to education* is shown in Appendix C.1. The case in which no document matches the keywords that have been entered has also been taken into account, to ensure no error message is displayed to the user. A screenshot of that case is available in Appendix C.2.

This chapter gave details on the dataset that was constructed and the platform that was implemented using the data collected. The next chapter discusses these results and their implications and highlights the limitations of the work that was carried out.

CHAPTER 5 DISCUSSION

This chapter summarises the work that was achieved during the project and highlights the limitations and possible future improvements.

5.1 Discussion

The proposed platform meets the functional requirements that were defined. It presents summary statistics on the collected human rights violations both generally, by country and by theme, and allows the user to filter incidents by keyword and by theme. The platform addresses the fragmentation of information that can make it difficult to identify patterns across reports. The combination of general statistics and filtering functions also enables users to examine data at different levels.

It is important to keep in mind that the data presented on the platform only represents the collected data. Just because a country does not have many reported incidents does not mean it actually has less incidents. A higher number of reported incidents cannot be interpretated as a higher actual number of incidents. Instead, the data on the platform should be interpreted as a representation of reported and documented incidents.

The dashboard, in particular provides a way of identifying patterns within the dataset that are not easily identify using other existing platforms on albinism in Africa. It allows users to explore the collected data on several layers. Nevertheless, reducing instances of human rights violations to data points can be risky, as there are human beings behind the data that was collected. Each incident has its own political, economic and social context and this should not be forgotten when using the platform and interpreting the data that it contains.

5.2 Limitations and future improvements

The first limitation of the project was that only a limited amount of data was collected. This was because the gemini API, that was used to extract excerpts from the document, has a limit of twenty requests per day. Therefore, only twenty documents could be analysed each day. The data collection process was carried out for approximately two weeks during the course of the project, to ensure enough time was allocated to finishing the platform building and writing the dissertation. To have more extensive data, the process can be repeated again for more time. Because, there were relatively few extracts in the dataset, some human rights theme, such as *Right to work*, were not represented extensively. Moreover, some African countries were not represented in the dataset, which slightly reduced the significance of the results.

Another limitation of the project was that not a lot of time was allocated to checking the data that was collected. Most users of the platform need reliable data for various purposes, which can not be fully assured by the platform at the moment. To improve the current work, more

time can be allocated to checking that the extracts are all relevant and that the themes and countries have been correctly allocated for each extract.

Finally, the platform can be improved during future works, by adding more interactivity. More information on each country can be shown to the user when they hover over it, or a new country specific dashboard can be opened upon clicking on each country. The process was started towards the end of the platform building process, but was abandoned to be able to focus on the testing phase, as a first finished and satisfying version of the platform had to be tested.

CHAPTER 6 CONCLUSION

The aim of the project was to build a human rights observatory for persons with albinism in Africa, using automated processes to collect documents available online and natural language processing methods to analyse them. The purpose of the platform was to display cases of human rights violations against persons with albinism in Africa

An automated data collection pipeline was built and implemented, to construct a dataset of more than 500 rows. More than 100 documents were analysed to extract meaningful and useful sentences on human rights violations of persons with albinism in Africa. Zero-shot classification and named-entity recognition were then used to assign, respectively, human right themes and countries to each extract. Once the data had been cleaned, the platform was built using Streamlit. The main features implemented in the platform were the interactive map and the search engine. All functions integrated in the platform and its general functioning were then evaluated using a questionnaire. The first part of the questionnaire was made off task-specific questions while the second used the system useability scale (SUS). It was then concluded that the platform performed well and in accordance with the requirements that were defined.

The platform addressed on of the main limitations of existing works on human rights of persons with albinism. It provides an interactive environment to explore available data on the topic. Overall, the project demonstrated that automated processes could be used to build a human rights platform.

Further work could be conducted on the platform, both to improve the reliability of the data and to increase the number of features available. To improve the reliability of the data, further work can include manually reviewing the excerpts that were extracted from documents, to ensure that each one mentions a human right violation and that no incident is mentioned twice. Moreover, adding more summary statistics for each country can be considered. Instead of a simple hovering function, the user could click on one country and see a new tab that is specific to this country.

Despite these limitations, the platform made it easier to compare and explore available and documents instances of human rights violations against persons with albinism in Africa.

BIBLIOGRAPHY

Al-Ababneh, H.A. *et al.* (2026) 'The Role of Functional and Non-functional Requirements in Creating a Quality Application', in K.F. Amawi (ed.) *AI-Driven Healthcare: Automation and Robotics*. Cham: Springer Nature Switzerland, pp. 1091–1105. Available at: https://doi.org/10.1007/978-3-032-15739-3_80.

'Albinism News & Stories' (2026) *Under the Same Sun*, 16 April. Available at: https://www.underthesamesun.com/news-stories/ (Accessed: 5 August 2026).

Bates, M. (1995) 'Models of natural language understanding.', *Proceedings of the National Academy of Sciences*, 92(22), pp. 9977–9982. Available at: https://doi.org/10.1073/pnas.92.22.9977.

BERT 101 - State Of The Art NLP Model Explained (2024). Available at: https://huggingface.co/blog/bert-101 (Accessed: 6 August 2026).

Cambria, E. (2025) 'Natural Language Understanding & AI', in E. Cambria (ed.) *Understanding Natural Language Understanding*. Cham: Springer Nature Switzerland, pp. 1–39. Available at: https://doi.org/10.1007/978-3-031-73974-3_1.

Convention on the Rights of Persons with Disabilities (2018) *Y v. United Republic of Tanzania*. CRPD/C/20/D/23/2014. OHCHR. Available at: https://juris.ohchr.org/casedetails/2511/en-US.

Dumouchel, S. *et al.* (2020) 'GOTRIPLE: A User-Centric Process to Develop a Discovery Platform', *Information*, 11(12), p. 563. Available at: https://doi.org/10.3390/info11120563.

Durojaye, E. and Nabaneh, S. (2019) 'HUMAN RIGHTS AND ACCESS TO HEALTHCARE FOR PERSONS WITH ALBINISM IN AFRICA', *African Disability Rights Yearbook*, 7. Available at: https://doi.org/10.29053/2413-7138/2019/v7a1.

Dykes, T. (2025) *Personas Make Users Memorable*, *Nielsen Norman Group*. Available at: https://www.nngroup.com/articles/persona/ (Accessed: 13 August 2026).

Ero, I. (ed.) (2016) *Enjoyment of human rights by persons with albinism: note*. New York: UN. Available at: https://digitallibrary.un.org/record/838972 (Accessed: 5 August 2026).

Ero, I. *et al.* (2019) 'PEOPLE WITH ALBINISM WORLDWIDE'.

Exa Search API - Exa (no date) *Exa Docs*. Available at: https://exa.ai/docs/reference/search-api-guide (Accessed: 12 August 2026).

Farivar, K. (2025) 'Semantic Search for Information Retrieval'. arXiv. Available at: https://doi.org/10.48550/arXiv.2508.17694.

Gabryel, M., Kocić, M. and Gabryel, A. (2026) 'Hybrid Retrieval in RAG: A Comparison of Semantic, Lexical and Reranking Methods', in L. Rutkowski et al. (eds) *Artificial Intelligence and Soft Computing*. Cham: Springer Nature Switzerland, pp. 86–93. Available at: https://doi.org/10.1007/978-3-032-03711-4_8.

Gardazi, N.M. *et al.* (2025) 'BERT applications in natural language processing: a review', *Artificial Intelligence Review*, 58(6), p. 166. Available at: https://doi.org/10.1007/s10462-025-11162-5.

Gemini API (no date) *Google AI for Developers*. Available at: https://ai.google.dev/gemini-api/docs (Accessed: 12 August 2026).

Global Albinism Alliance (no date) *Albinism Human Rights News*, *Global Albinism Alliance*. Available at: https://www.albinismalliance.org/human-rights-news (Accessed: 5 August 2026).

Han, S.C. *et al.* (2025) 'Applications and Case Studies in Natural Language Understanding', in S.C. Han et al. (eds) *Natural Language Understanding in Conversational AI with Deep Learning*. Cham: Springer Nature Switzerland, pp. 129–145. Available at: https://doi.org/10.1007/978-3-031-74364-1_6.

He, P. *et al.* (2021) 'DeBERTa: Decoding-enhanced BERT with Disentangled Attention'. arXiv. Available at: https://doi.org/10.48550/arXiv.2006.03654.

Kalyani, P. (2025) *Comprehensive Overview of Named Entity Recognition: Models, Domain-Specific Applications and Challenges*. Available at: https://arxiv.org/html/2309.14084 (Accessed: 6 August 2026).

Khurana, D. *et al.* (2023) 'Natural language processing: state of the art, current trends and challenges', *Multimedia Tools and Applications*, 82(3), pp. 3713–3744. Available at: https://doi.org/10.1007/s11042-022-13428-4.

Laurer, M. *et al.* (2024) 'Building Efficient Universal Classifiers with Natural Language Inference'. arXiv. Available at: https://doi.org/10.48550/arXiv.2312.17543.

Lund, P.M. *et al.* (1997) 'Oculocutaneous albinism in an isolated Tonga community in Zimbabwe.', *Journal of Medical Genetics*, 34(9), pp. 733–735. Available at: https://doi.org/10.1136/jmg.34.9.733.

Muscati, S. (no date) *HRW: From Cradle to Grave: Discrimination and Barriers to Education for Persons with Albinism in Tete Province, Mozambique*. Available at: https://www.hrw.org/video-photos/interactive/2019/06/13/cradle-grave (Accessed: 5 August 2026).

Plan international (no date) *Girls' rights platform*. Available at: https://www.girlsrightsplatform.org/library/ (Accessed: 6 August 2026).

Rayhan, A., Kinzler, R. and Rayhan, R. (2023) 'NATURAL LANGUAGE PROCESSING: TRANSFORMING HOW MACHINES UNDERSTAND HUMAN LANGUAGE', 26 August. Available at: https://doi.org/10.13140/RG.2.2.34900.99200.

Renjit, S. and Idicula, S.M. (2024) 'A Study of the State of the Art Approaches and Datasets for Multilingual Natural Language Inference', *Neural Processing Letters*, 56(6), p. 243. Available at: https://doi.org/10.1007/s11063-024-11673-2.

Saravanos, A. (2025) 'A Brief History of the Waterfall Model: Past, Present, and Future', *Proceedings of the 2025 18th International Conference on Computer Science and Information Technology*. *ICCSIT 2025: The 18th International Conference on Computer*

Science and Information Technology, Paris France: ACM, pp. 138–143. Available at: https://doi.org/10.1145/3783862.3783879.

Sauro, J. (2011) 'Measuring Usability with the System Usability Scale (SUS) – MeasuringU'. Available at: https://measuringu.com/sus/ (Accessed: 12 August 2026).

Standing Voice (no date) *Exposure*. Available at: https://news.standingvoice.org/ (Accessed: 5 August 2026).

Streamlit • A faster way to build and share data apps (2021). Available at: https://streamlit.io/ (Accessed: 12 August 2026).

Supriyono *et al.* (2024) 'Advancements in natural language processing: Implications, challenges, and future directions', *Telematics and Informatics Reports*, 16, p. 100173. Available at: https://doi.org/10.1016/j.teler.2024.100173.

Tambala-Kaliati, T., Adomako, E.B. and Frimpong-Manso, K. (2021) 'Living with albinism in an African community: exploring the challenges of persons with albinism in Lilongwe District, Malawi', *Heliyon*, 7(5), p. e07034. Available at: https://doi.org/10.1016/j.heliyon.2021.e07034.

United Nations (1948) 'Universal Declaration of Human Rights'. United Nations. Available at: https://www.un.org/en/about-us/universal-declaration-of-human-rights (Accessed: 18 August 2026).

United Nations (1966) 'International Covenant on Economic, Social and Cultural Rights'. Available at: https://www.ohchr.org/en/instruments-mechanisms/instruments/international-covenant-economic-social-and-cultural-rights (Accessed: 18 August 2026).

United Nations (2006) 'Convention on the Rights of Persons with Disabilities'. Available at: https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities (Accessed: 18 August 2026).

United Nations (no date) *Meet the Champions of the Albinism Cause - United Nations Human Rights*. Available at: http://albinism.ohchr.org/ (Accessed: 6 August 2026).

What is Zero-Shot Classification? - Hugging Face (2023). Available at: https://huggingface.co/tasks/zero-shot-classification (Accessed: 6 August 2026).

World Health Organisation (2026) *Nigeria [country overview]*, *datadot*. Available at: https://data.who.int/countries/566 (Accessed: 18 August 2026).

Wright, C.Y., Norval, M. and Hertle, R.W. (2015) 'Oculocutaneous Albinism in Sub-Saharan Africa: Adverse Sun-Associated Health Effects and Photoprotection', *Photochemistry and Photobiology*, 91(1), pp. 27–32. Available at: https://doi.org/10.1111/php.12359.

Zawanah (2025) 'BM25 Explained: The Classic Algorithm that Still Powers Search Today', *Medium*, 3 October. Available at: https://medium.com/@zawanah/bm25-explained-the-classic-algorithm-that-still-powers-search-today-865351fce9aa (Accessed: 7 August 2026).

APPENDIX

A Code files – Data collection

A.1 Data collection function

```
def collection(query, n_answers=20, url_seen=[]):
    '''
    input:
    query: str, keywords used for the search
    n_answers: int, number of documents to return
    url_seen: list, list of urls that have already been treated

    output:
    pandas dataframe with details on the documents analysed
    '''

    # step 0: creating an empty dataframe to store the results
    data = pd.DataFrame(columns=["url", "title", "authors", "date", "extract",
"country", "theme"])

    # step 1: searching for documents on internet
    print("-"*20)
    print("searching internet with query: ", query)

    results = exa.search(
        query=query,
        num_results=n_answers,
        user_location="GB",
        exclude_domains=["wikipedia.org"],
        category="news"
    )
    print("done")

    # step 2: getting a list of the new urls collected

    urls = []
    for i in range(len(results.results)):
        url = results.results[i].url

        if url not in url_seen: # checking against the urls that have already
been treated
            urls.append(url)

    print("Number of new urls collected: ", len(urls))
    print("-"*20)
```

```
    # step 3: for each url, get the content, pass it through the LLM

    i = 1 # initialising counter for printing

    for url in urls:

        print(f"Analysing url {i} of {len(urls)}")

        # get the content of the url
        content = exa.get_contents(
            urls=[url],
            text=True
        )
        text = content.results[0].text

        print("content collected")

        # using the llm to extract the information

        interaction = client.interactions.create(
            model="gemini-3.5-flash",
            input=f"Return the detailed instances of human rights violation
against a specific person with albinism that happened in Africa: {text}",
            response_format={
                "type": "text",
                "mime_type": "application/json",
                "schema": Violations.model_json_schema()
            }
        )

        print("gemini successful")

        llm_output = interaction.output_text
        output = Violations.model_validate_json(llm_output)

        # adding the new extracts to the results dataframe
        for extract in output.violation_sentences:
            new_row = pd.DataFrame({
                "url":[url],
                "title":[output.title],
                "authors":[output.author],
                "date":[output.date],
                "extract":[extract],
                "country":None,
                "theme":None

            })
```

```
            data = pd.concat([data, new_row], ignore_index=True)

        print(f"{output.n_violations} were added to the dataframe")

        i += 1

        if i > 20:
            break

        # sleeping for 1 minute to make sure to not exceed the max number of
requests per minute
        sleep(60)

    return data
```

A.2 Analysis function

```
def analyse(data, themes=default_themes, map_dic={}):
    '''
    input:
    data: pandas dataframe, containing the extracts to analyse
    themes: list of default themes to classify the extract in
    map_dic: dictionnary of places that have already been mapped

    output:
    new_data: pandas dataframe, with the themes and country columns filled
    new_map_dic: new mapping dictionnary
    '''


    # step 1: creating a copy of the dataframe and of the mapping dictionnary
    new_data = data.copy()
    new_map_dic = map_dic.copy()

    # resetting the index of the dataframe
    new_data = new_data.reset_index()

    # step 2: adding the theme and places mentionned for each line
    n_lines = len(new_data) # number of lines to analyse

    for i in range(n_lines):

        # only analyse if there is nothing in the country and theme column
        if pd.isna(new_data["country"].loc[i]) and
pd.isna(new_data["theme"].loc[i]):
            print("-"*20)
            print(f"analysing line {i}")
            print("-"*20)
```

```
            extract = new_data["extract"].loc[i]
            title = new_data["title"].loc[i]

            # step 2.a: adding the theme(s) of the extract
            id_themes = classification_function(extract, themes=themes)

            new_data.at[i, "theme"] = id_themes

            #step 2.b: adding the countries of the extract
            print("Extracting places names")
            places = places_recognition(extract)

            if len(places) == 0: # if no place has been identified in the
extract, test with the title
                places = places_recognition(title)

            print(f"{len(places)} places identified, extracting country
names")
            # initialising the country list
            countries = []

            for place in places:
                country, new_map_dic = map_place_to_country(place,
new_map_dic)
                if country not in countries:
                    countries.append(country)
                time.sleep(15) # making sure the quotas are not reached

            new_data.at[i, "country"] = countries
            print(f"found {len(countries)} countries")

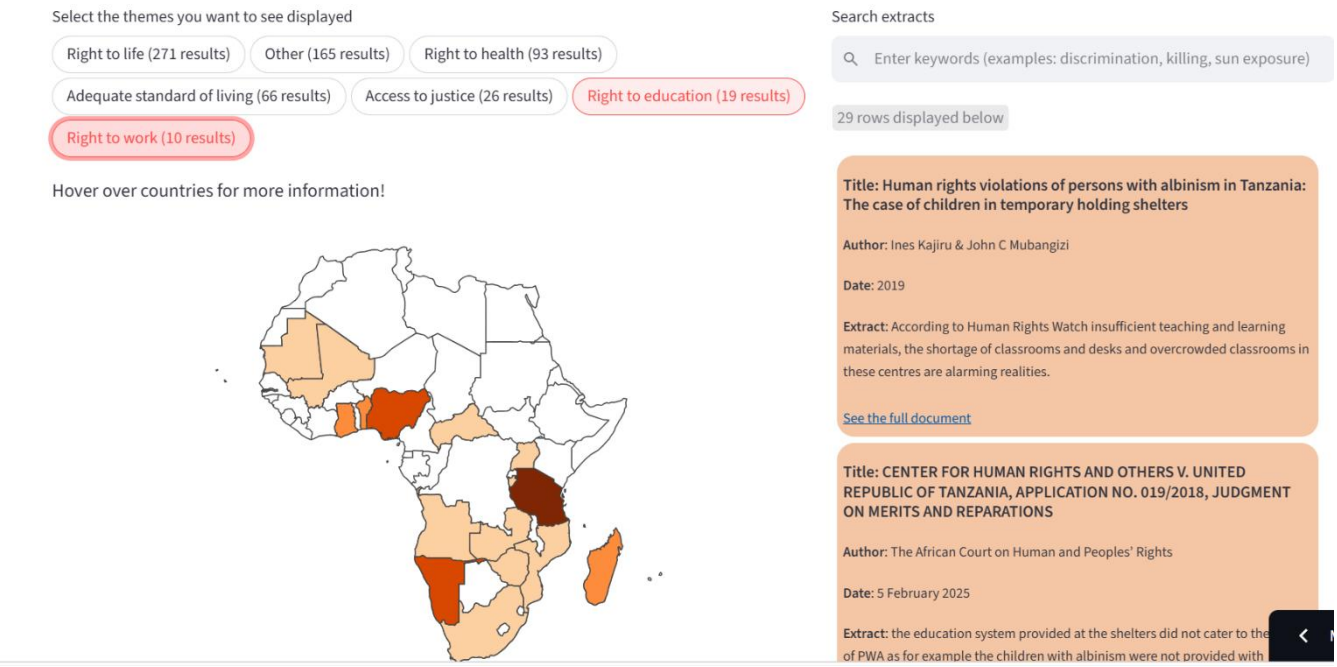
    return new_data, new_map_dic
```

B Code files – Platform building

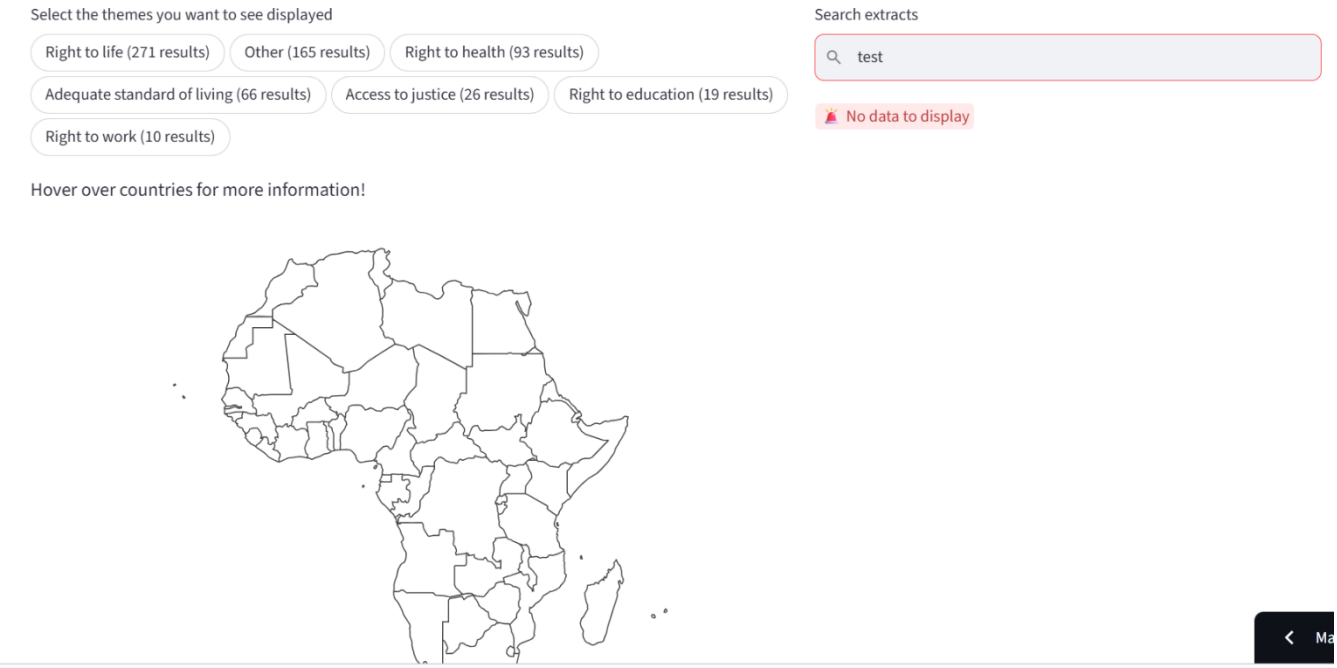
The code for the platform is available in the following public Github repository: eliiiiiiiiise/dissertation

C Additional wireframes of the final platform

C.1 Selecting multiple themes



C.2 No document matches the keywords



D Ethics documents

D.1 Participant information sheet

You are being invited to participate in a study run by the Dept of Management Science of Strathclyde University to evaluate an online platform that was built to visualise recorded occurrences of human rights violations on people with albinism in Africa. The study is run by Elise L'Hospitalier, as part of their Dissertation in Data Analytics. This information sheet is provided to you so that you can decide whether or not you wish to participate.

The aim of the study is to collect and analyse publicly available online resources relating to human rights issues affecting people with albinism in Africa, and to present these resources through an online observatory platform. As part of this project, participants are invited to evaluate the effectiveness and usability of the platform.

If you agree to participate, you will be asked to complete a task on the platform to find information relating to human rights issues affecting people with albinism in Africa, followed by a short questionnaire evaluating the usability of the platform – this should take no more than 20 minutes. You will then be invited to take part in a brief exit interview to capture your general experience using the platform.

Please be aware that as part of this study you may be exposed to content relating to human rights violations against people with albinism, which may be distressing. You are under no obligation to continue participation at any point and support resources will be made available should you experience any discomfort.

Your participation is voluntary, and you may decide to withdraw at any time without giving a reason. In this case any data you have given will be withdrawn from the study and destroyed. The numerical data collected will be used to assess the platform's effectiveness and identify areas of improvement. Aggregated data across all respondents may be used in the final report. Short quotes from exit interviews may be used in the final report, but these shall be used in such a way that the source is not identifiable.

The data collected will be held on the University server on the researcher's password protected account. This study is being conducted by Elise L'Hospitalier, a MSc Data Analytics student at Strathclyde University, as part of their final year project.

The work is supervised by Dr Catherine Chavula, a member of staff at the Dept of Computer and Information Sciences and has been approved by the Management Sciences Department Ethics Committee.

If you have any questions about the study, please feel free to contact us:

Elise L'Hospitalier, Email: elise.l-hospitalier.2025@uni.strath.ac.uk

Dr Catherine Chavula, Email: catherine.chavula@strath.ac.uk

Support resources:

If you experience any emotional distress during or after your participation in this study, please do not hesitate to use the following resources:

University of Strathclyde Students:

Spectrum Life: confidential counselling and wellbeing support available 24 hours a day, 7 days a week

Freephone (UK): 0800 031 8227

General public (UK):

NHS 24: Free medical and mental health advice available 24 hours a day

Phone: 111

D.2 Participant consent form

I have read and understood the participant information sheet and agree to take part in this study, including completing a task on the platform, completing a usability questionnaire and exit interview. I understand that I may be exposed to content relating to human rights violations against people with albinism, and that I may withdraw from participation at any time without consequence.

I agree to the information and data I provide during this study being collected, stored, and used solely for the purposes specified in the participant information sheet.